汽车电子与控制
Automotive Electronics and Control Systems (AECS)

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汽车安全与节能国家重点实验室

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第二章：汽车电控系统数字核心（单片机）
Digital Core of AECS (Microcontroller)

§目录 / Index

1 TC系列MCU结构与编程

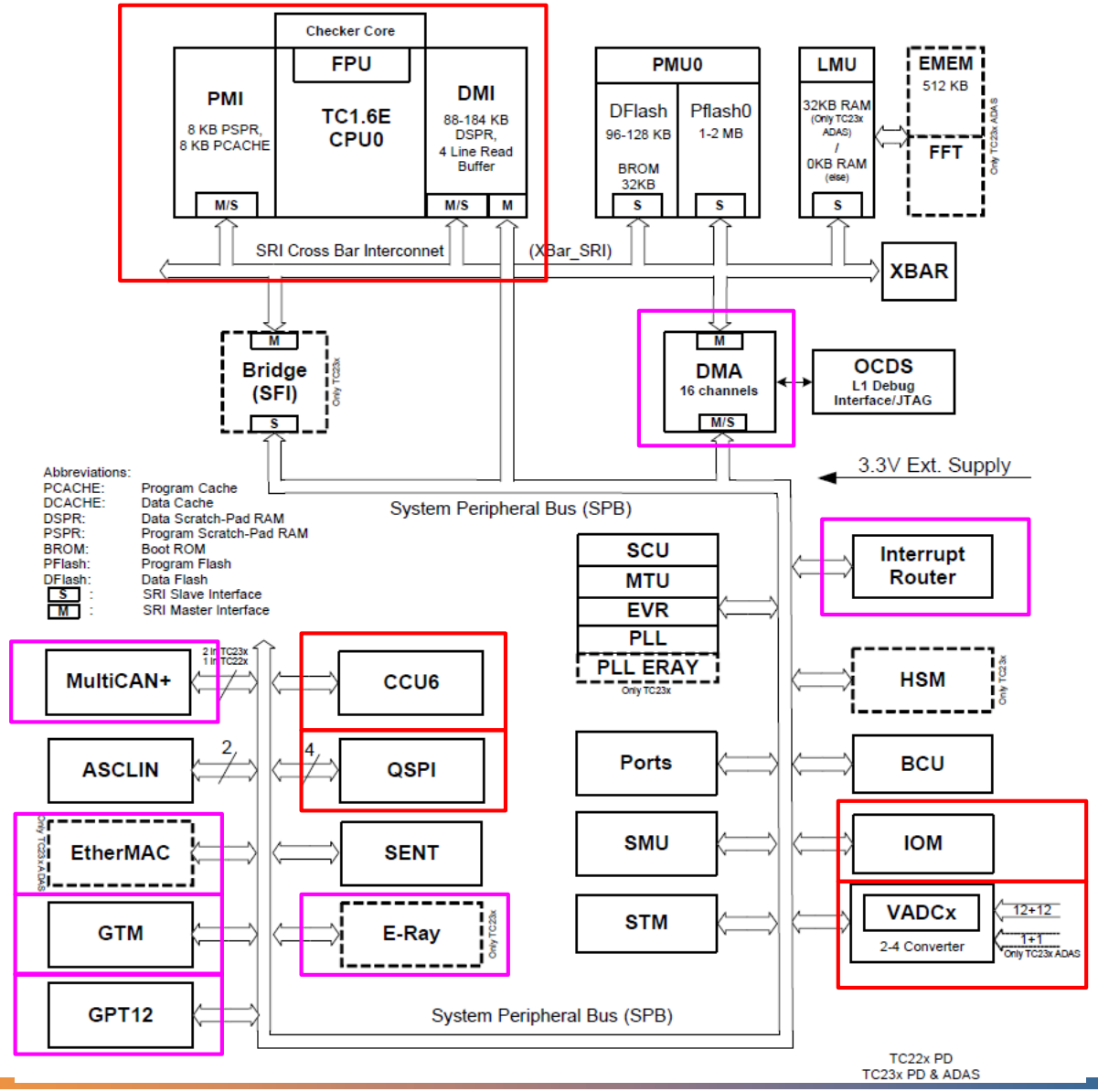
2 TC系列MCU指令系统

3 TC系列MCU的CPU与外围电路

4 TC系列MCU的周边模块

本科阶段应掌握的内容

研究生阶段应掌握的内容



AURIX Training

4.1 SAR concepts

Please note, that outlined features and target parameters are preliminary and might change without notice

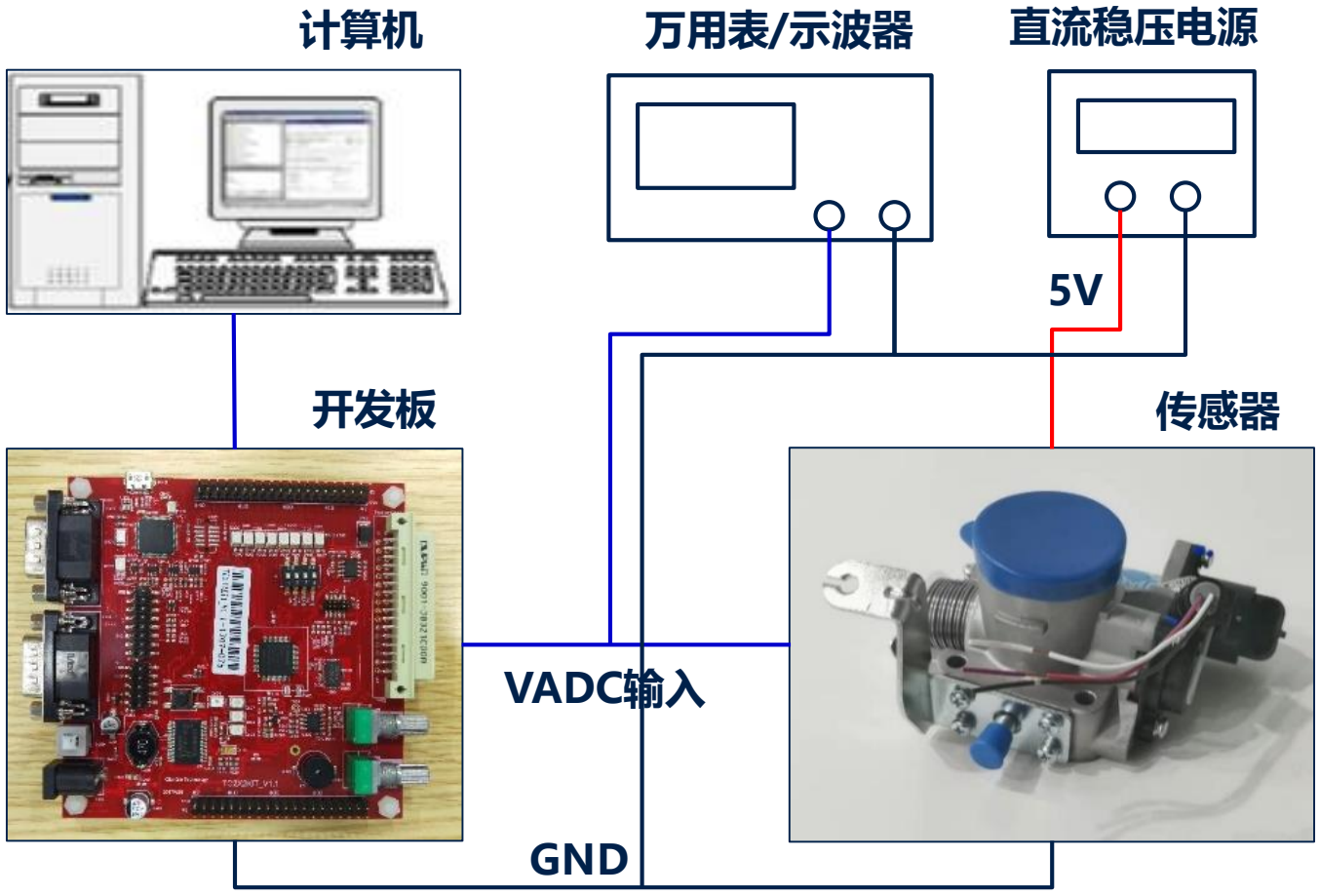
IFAG ATV MC ACE ATV
Klaus Scheibert
Principal Powertrain
October 19th, 2011, V1.0

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Table of contents

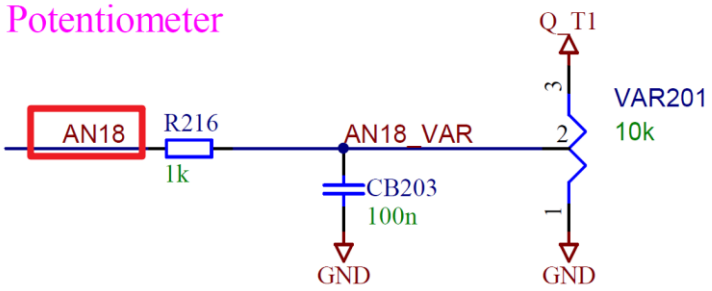
- 4.1.1 An example
- 4.1.2 Overview
- 4.1.3 Module Activation & Control
- 4.1.4 Conversion Request
- 4.1.5 Input channel configuration
- 4.1.6 Conversion timing
- 4.1.7 Conversion Result Handling
- 4.1.8 Synchronization of conversion
- 4.1.9 Safety Feature
- 4.1.10 External Multiplexer control
- 4.1.11 Service Request & Trigger/Gate

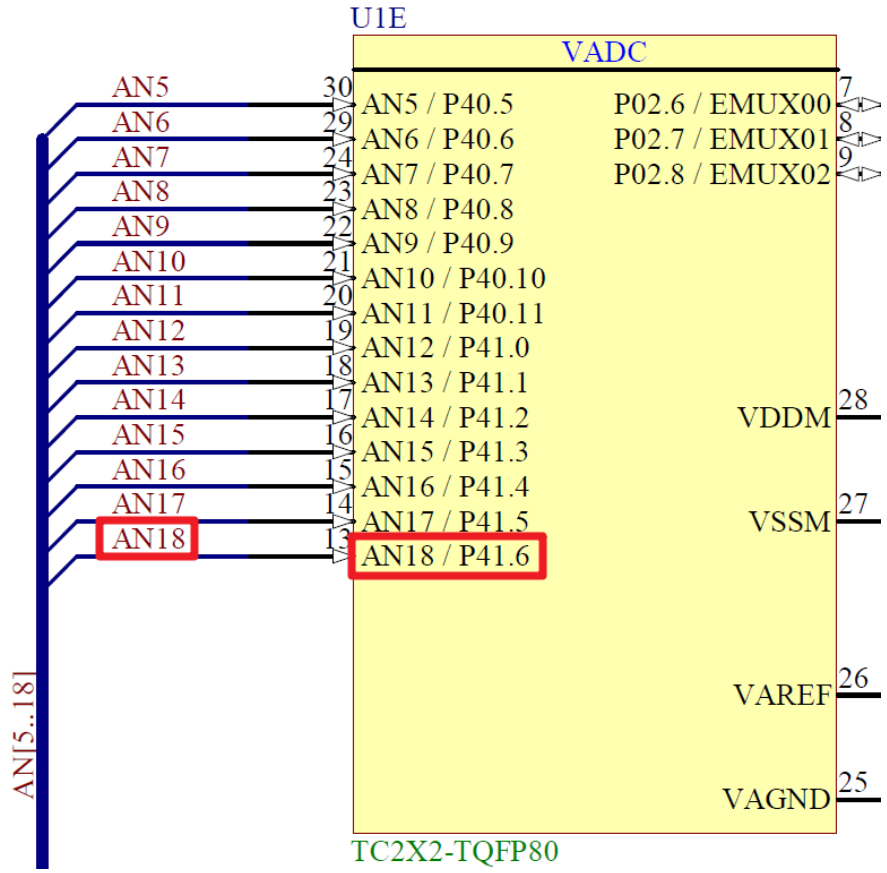
4.1.1 An example



硬件怎么连?

4.1.1 An example





参考文件：

TC212_datasheet
TC212_usermanual

TC21xTC22x_DS_v1_0.pdf
tc23x_tc22x_um_v1.1.pdf

13	P41.6	I	S / VDDM	General-purpose input
	VADCG1_6	AI		VADC input channel 6 of group 1

单片机管脚怎么对应?

4.1.1 An example

AD采样程序结构

定义VADC模块初始化函数：

- 模块使能
- AD转换设置
- 关闭寄存器写入权限
- 配置对应I/O端口为输入

```
#include "SysSe/Bsp/Bsp.h"
#include "IfxScuWdt.h"
#include "Ifx_reg.h"

void AD_Init(void)
{
    MODULE_VADC.CLC.B.DISR = 0;                    //VADC模块使能
    while(MODULE_VADC.CLC.B.DISS!=0)               //由于模块使能实现由一定延误，所以为了保证紧
接着的代码能顺利执行，直至DISS=0表明模块使能完成后才脱离循环
    {
        ;
    }
    ……//此处省略了部分寄存器设置代码，具体可见例程
    while(MODULE_VADC.G[1].ARBCFG.B.CAL!=0)    //等待初始标定完成
    {
        ;
    }
    MODULE_VADC.G[1].ASMR.B.LDEV=1;      //generate a load event, AD模块开始自动扫描

    //关闭所有寄存器的写入权限
    MODULE_VADC.ACCPROT0.U = 0x800F800F;
    MODULE_VADC.ACCPROT1.U = 0x000F800F;
    MODULE_P41.IOCR4.B.PC6=0;    //P41.6引脚配置为输入

}
```

寄存器怎么设置？如何通过代码实现？

4.1.1 An example

```
int core0_main(void)
{

    IfxCpu_disableInterrupts();      //禁止中断

    //初始化时钟系统，fCPU=100MHz
    IfxScuCcu_Config scuccuconfig;
    IfxScuCcu_initConfig(&scuccuconfig);
    IfxScuCcu_init(&scuccuconfig);

    //关闭CPU看门狗和Safety看门狗，如果开启看门狗，需要定期喂狗
    IfxScuWdt_disableCpuWatchdog(IfxScuWdt_getCpuWatchdogPassword());
    IfxScuWdt_disableSafetyWatchdog(IfxScuWdt_getSafetyWatchdogPassword());

    AD_Init();


    while (1)
    {
        k += 1;
        res = MODULE_VADC.G[1].RES[6].B.RESULT;
        delay(0.1);
    }
    return 0;
}
```

主程序：

- **禁止中断**
- **初始化时钟**
- **配置看门狗**
- **VADC模块初始化**
- **死循环**
  - **读取VADC结果寄存器，获取AD采样结果**

4.1.1 An example

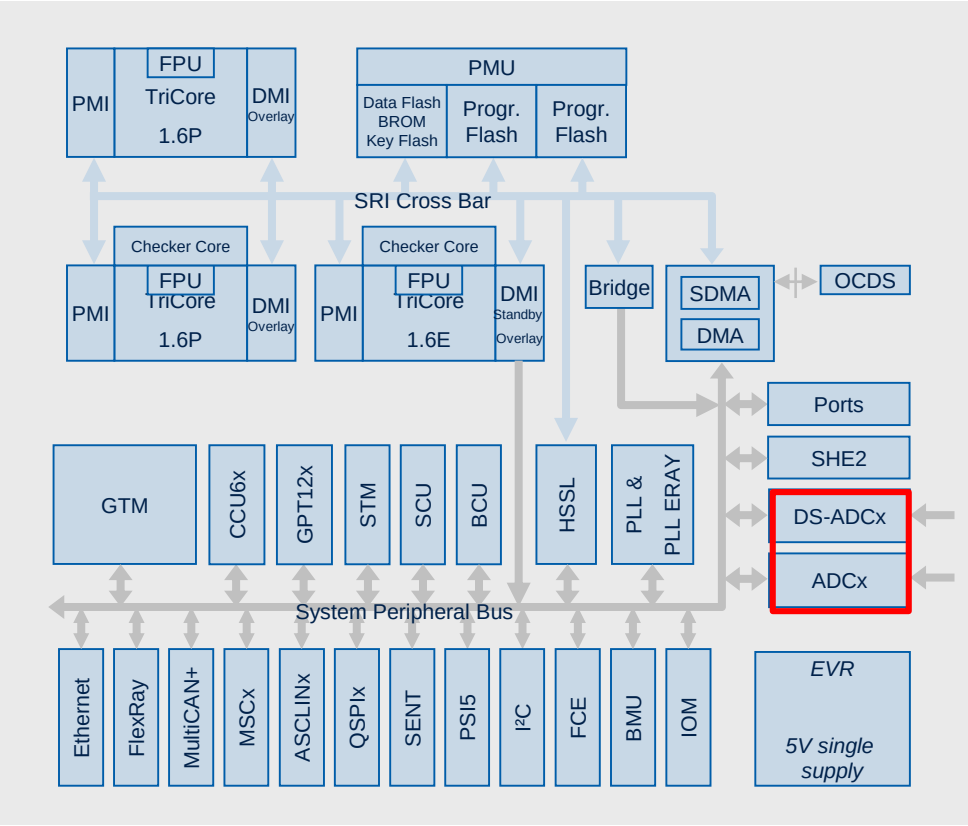
VADC模块初始化需要配置的寄存器：

- Clock Control Register(CLC)-时钟使能
- Access Protection Register (ACCPROT0，ACCPROT1)
- GxARBCFG-AD模块激活
- Global Configuration Register (GLOBCFG)-时钟分频
- GxARBPR (x = 0 – 3)-优先级
- GxASMR (x = 0 – 3)-AutoScan模式
- GxICLASS0 (x = 0 – 3)-输入通道采样时间、转换字长
- GxCHCTRy (x = 0 – 3)-输入通道成组方式、结果存放
- GxASSEL (x = 0 – 3)-通道加入扫描序列
- GxRESy (y=0~15)-读取转换结果

x: 模拟量输入通道的Group序号，0~3
y: 模拟量输入通道的Channel序号

掌握最基本的：Autoscan采样方式

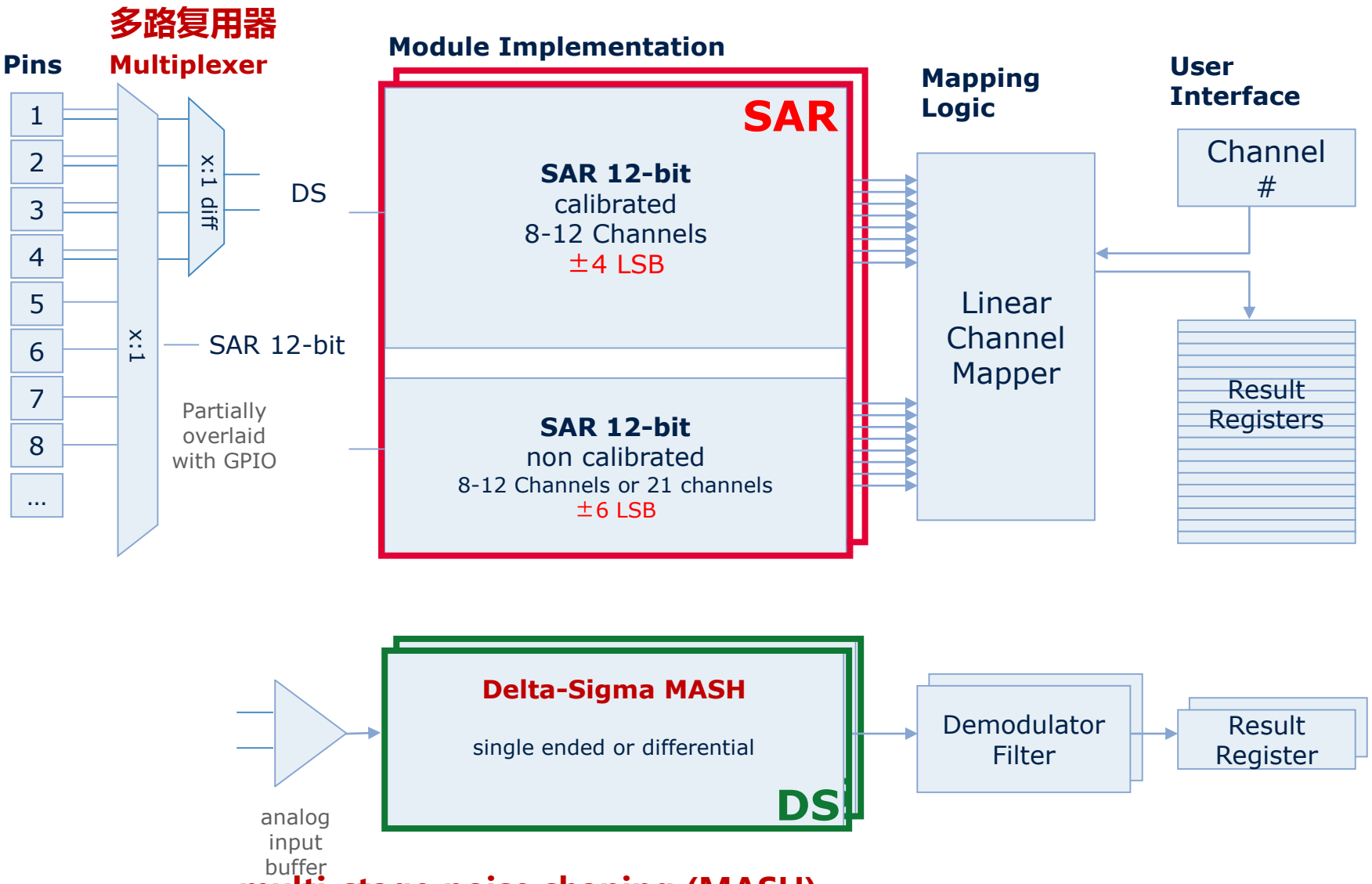
4.1.2 Overview
AURIX Differentiation Strategy



- **ADC**
  - **Delta-sigma ADCs** with up to 13-bit accuracy
  - 12-bit **SAR ADCs** with conversion time <1 µs
  - Multiple SAR modules allow high number of simultaneous conversions
  - Flexible source selection and arbitration
  - Powerful result handling
  - Flexible service request generation
  - Built-in safety features
  - > **80 ADC** channels on some devices

ADC: Anolog-to-Digital Converter, VADC: Versatile ADC, DS-ADC: Delta-sigma ADC
SAR ADC: Successive Approximation Register ADC 逐次逼近型ADC

4.1.2 Overview
ADC/DS Implementation Concept



multi-stage noise shaping (MASH)

Single ended input: 单端输入；double ended differential input: 差分输入

4.1.2 Overview
ADC/DS Device Configurations

AURIX Device Number		9x Series	7x Series	6x Series	4x Series
Analog inputs [dedicated / overlaid with GPIO]		BGA516: 72/12 BGA416: 48/12 BGA292: 48/12	BGA292: 48/12 QFP176: 36/12	BGA292: **44**/12 QFP176: 36/12 QFP144: 28/10	QFP144: 28/**12** QFP100:16/**9**
	calibra-ted	4x 8 ch	4x 8 ch	**SAR0 - 16 ch** **SAR1&2 – 14 ch**	-
	Non calibra-ted	5 x 8 ch / 2 x 6 ch	2 x 8 ch / 2 x 6 ch	1x 12 ch	2x 16 ch 1x 12 ch
Delta-Sigma modules (Twin 1-1 MASH)		10x 1 diff ch (≥74-83dB)	6x 1 diff ch (≥74-83dB)	**3x 1 diff ch (≥74-83dB)**	**2x 1 diff ch (≥74-83dB)**
Multiplexer Granularity		1x4:1 **4x2:1** **5x1:1**	1x4:1 (DS3) 1x2:1 (DS2) 4x1:1 (DS0,DS1, DS4, DS5)	**1 x 4:1 (DS1)** **1 x 2:1 (DS0)** **1 x 1:1 (DS2)**	1 x 2:1 (DS1) 1 x 1:1 (DS0)
	12-bit SAR	11	8	4	3
	DS1	10	6	3	2

Note: red items are under evaluation for open market

SAR specification

- the following 12-bit SAR configuration is assumed
  - calibrated **12-bit ADC** (TUE ± 4LSB → ENOB = 10 bit)
  - non calibrated **12-bit ADC** (TUE ± 6LSB) → ENOB = 9.5 bit

TUE: Total Unadjusted Error

ENOB: Effective Number of Bits

DS specification:

- **MASH spec values**
  - 74-80 dB for 100-30kHz signal input bandwidth
  - ENOB = 12 – 13

4.1.2 Overview
ADC/DS Device Configurations

AURIX Device		9x Series 150 Mhz – 1µs/ch	7x Series 150 Mhz – 1µs/ch	6x Series 80 Mhz – 1µs/ch	4x Series 80 Mhz – 1µs/ch
Analog inputs [dedicated / overlaid with GPIO]		BGA516: 72+12			
		BGA416: 48+12			
			BGA292: 48+12		
			QFP176: 36+12	QFP176: 38/12	
				QFP144: 30/10	QFP144: 30/10
					QFP100:16/9
SAR0		SH0/1/2/3 8ch each	SH0/1/2/3 8ch each	SH0/1 16ch each	SH0/1 16ch each
SAR1		SH4/5/6/7 8ch each	SH4/5/6/7 8ch each	SH2/3 16ch each	SH2/3 16ch each
SAR2		SH8/9/10 8ch each	n.a.	n.a.	n.a
Delta-Sigma modules		10x 1 diff ch	6x 1 diff ch	3x 1 diff ch	2x 1 diff ch
PDD/MUXDIAG/ALTREF		PDD-(AN7/AN35) MUX-(VADCGx.1/x.2) ALTREF(VADCGx.0) noALTREF(VADCG0.[4:7],AN35) 4ch DfT assumed	PDD-(AN7/AN35) MUXDIAG-(ADCxSHx.1/2) ALTREF(ADCxSHx.0) noALTREF(ADC0SH0.4-7, ADC1SH0.3) 4ch DfT assumed	PDD-(AN7/AN35) MUXDIAG-(ADCxSHx.1/2) ALTREF(ADCxSHx.0) noALTREF(ADC0SH0.4-7, ADC1SH0.3) 4ch DfT assumed	PDD-(AN7/AN35) MUXDIAG-(ADCxSHx.1/2) ALTREF(ADCxSHx.0) noALTREF(ADC0SH0.4-7, ADC1SH0.3) 4ch DfT assumed
Multiplexer Granularity		1x4:1(DS3), 4x2:1 (DS2,DS6,DS7,DS9), 5x1:1 (DS0,DS1, DS4, DS5,DS8)	1x4:1 (DS3) 1x2:1 (DS2) 4x1:1 (DS0,DS1, DS4, DS5)	1 x 4:1 (DS3) 1 x 1:1 (DS2) 1 x 2:1 (DS0)	1 x 2:1 (DS3) 1 x1:1 (DS0)
No of Modules	SH	11	8	4	4
	DS	10	6	3	2

4.1.2 Overview
AURIX TC212L ADC模块

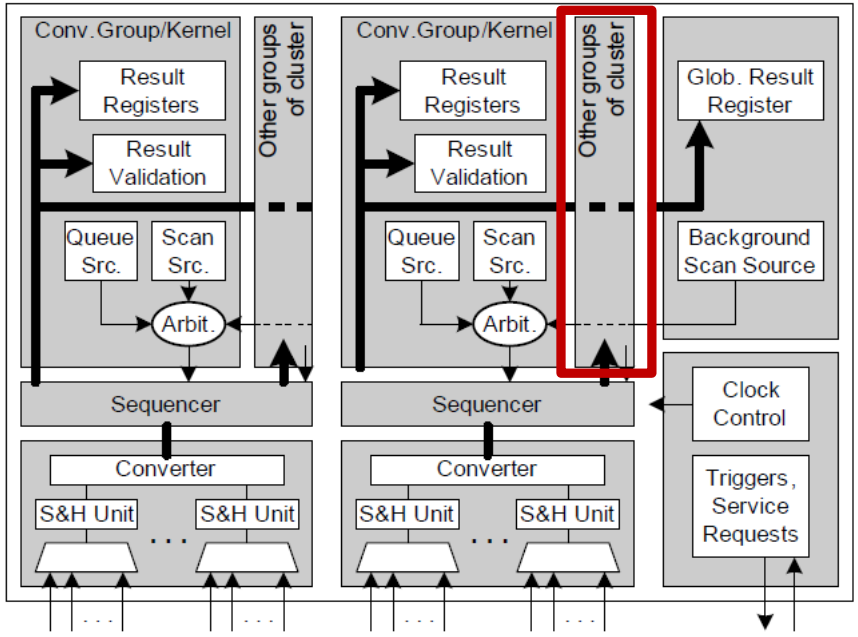
Feature		TC21x	TC22x	TC23x	TC23x ADAS
CPU Core	Type	TC1.6E			
	Cores / Checker Cores	1 / 1[1)]		1 / 1	
	Max. Freq.	133 MHz		200 MHz	
	FPU	yes			
	MPU	yes			
Program Flash	Size	512 Kbyte	1 Mbyte	2 Mbyte	
Data Flash	Size	64 Kbyte	96 Kbyte	128 Kbyte	
Cache	Instruction	8 Kbyte			
	Data	4 line read buffer			
SRAM	Size (DSPR/PSPR)	48 / 8 Kbyte	88 / 8 Kbyte	184 / 8 Kbyte	
	Size LMU	0 Kbyte			32 Kbyte
	Size EMEM	0 Kbyte			512 Kbyte
DMA	Channels	16			
ADC (SAR)	Voltage range	0 … 5.5 V (determined by ADC supply)			
	Inputs	24			
	Converters	2			4
	Resolution	12 Bits			
	Channels	12 + 12			12 + 12 + overlay 1 + 1

参考文件：
TC212_datasheet
TC212_usermanual

TC21xTC22x_DS_v1_0.pdf
tc23x_tc22x_um_v1.1.pdf

4.1.2 Overview AURIX SAR

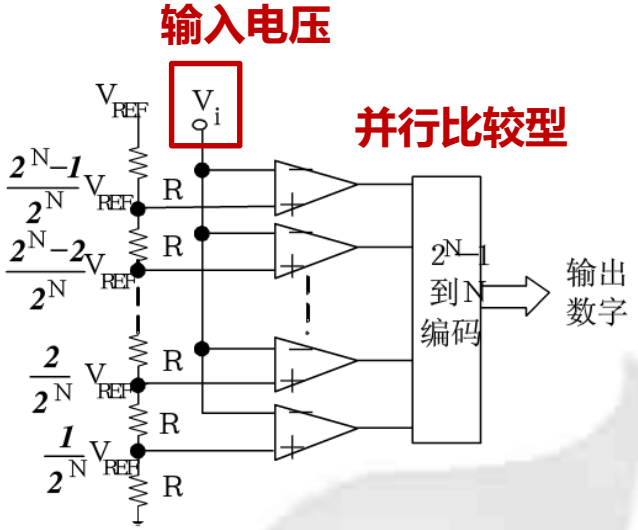
- VADC provides a series of analog input channels connected to a cluster of Analog/Digital Converters using the Successive Approximation Register (**SAR**) principle to convert analog input values (voltages) to discrete digital values
- VADC is based on **Sample&Hold** converters, arranged in **clusters**. Each cluster contains **4 Sample&Hold units** which share a common converter.
- Each converter of the ADC cluster can operate independent of the others, controlled by a **dedicated** set of registers and triggered by a dedicated **group request source**
- A **background request source** can access all analog input channels that are not assigned to any group request source
  - these conversions are executed with **low priority**
  - the background request source can, therefore, be regarded as an additional (virtual) background converter

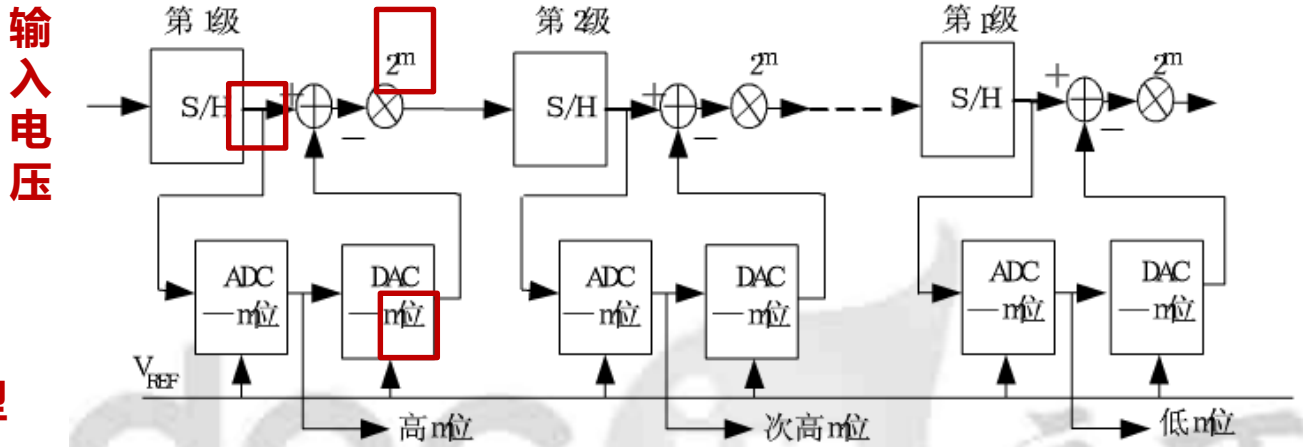


AD转换器的类型

并行比较型：转换速率极高，电路规模极大
逐次逼近型：转换速率较高、电路规模中等
其他：

- 串并行型
- 积分型
- Σ-Δ调制型
- 电容阵列逐次比较型
- 压频变换型





逐次逼近型

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AD转换器的主要技术指标

- 分辨率：n位输出的A/D转换器能区分2^n个不同等级的输入模拟电压，分辨率为$V_{ref}/ 2^n$。
- 转换时间：从转换控制信号到来开始，到输出端得到稳定的数字信号所经过的时间，并行比较型ADC转换时间较短。
- 量化误差：A/D转换器实际输出的数字量和理论上的输出数字量之间的差别，常用最低有效位（LSB：Least Significant Bit）的倍数表示。例如给出相对误差不大于±LSB/2，这就表明实际输出的数字量和理论上应得到的输出数字量之间的误差小于最低位的半个Bit。

4.1.2 Overview
Feature Set (1)

- Input voltage range from **0 V up to analog supply voltage**
- Standard (*V*AREF) and alternate (CH0) reference voltage source selectable for each channel to support ratio-metric measurements and different signal scales
- Up to 11 independent converters
- Up to 8 analog input channels for each converter
- The module can handle **16 channels**
  - actual number depends on product
- External analog multiplexer control, including adjusted sample time and scan support
- Flexible source selection and **arbitration（仲裁）**
  - Programmable arbitrary conversion sequence (single or repeated)
  - Single-channel conversion (single or repeated)
  - Configurable **auto scan** conversions (single or repeated) in the background
  - Conversions **triggered by software, timer events, or external events**
  - **Cancel-inject-restart mode** for reduced conversion delay on priority channels

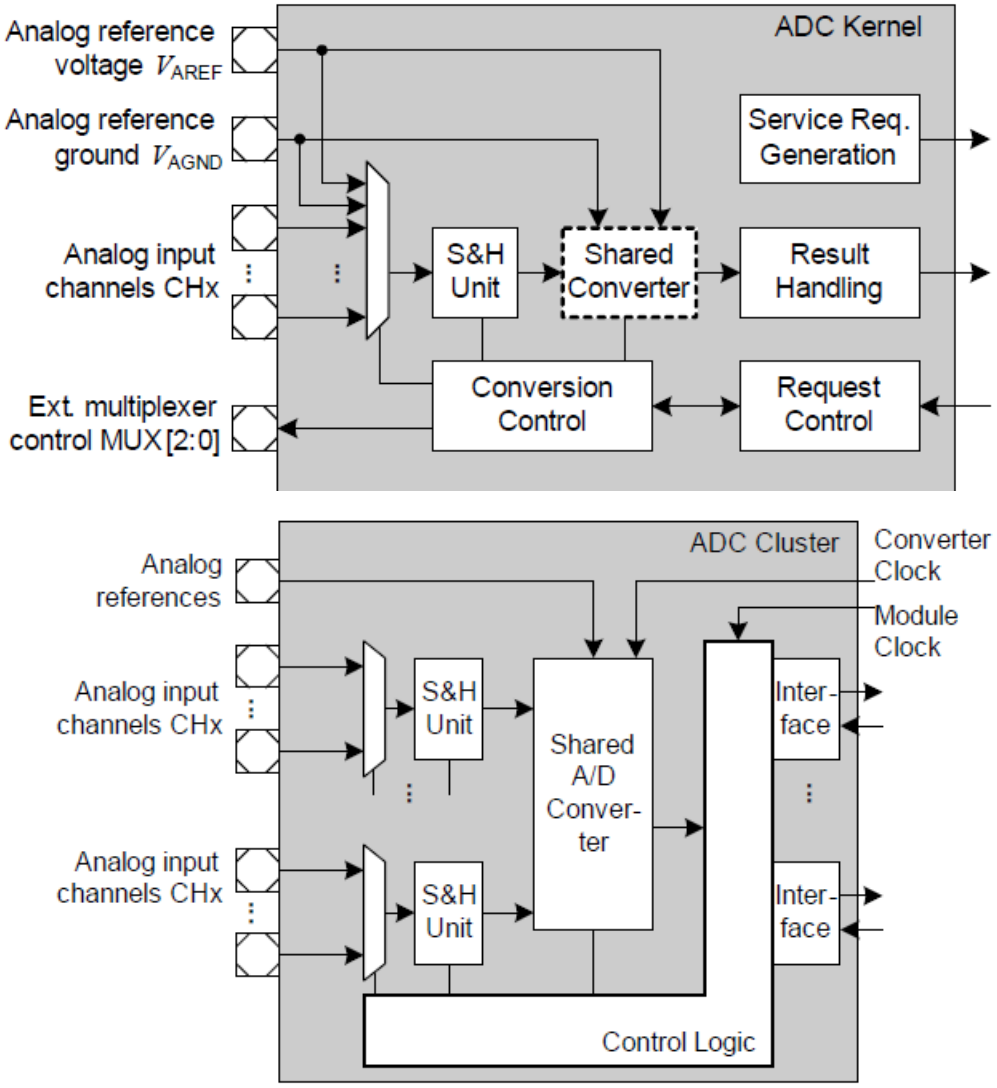
4.1.2 Overview
Feature Set (2)

- Powerful result handling
  - Selectable result width of **8/10/12** bits
  - Independent result registers
  - Configurable limit checking against programmable border values
  - Data rate reduction through adding a selectable number of conversion results
  - **FIR/IIR** filter with selectable coefficients
- Flexible service request generation based on selectable events
- Built-in safety features
  - Configurable register access protection supports safety applications
  - Broken wire detection with programmable default levels
  - Multiplexer test mode to verify signal path integrity
- Support of suspend and power saving modes
- Conversion speed and sample time adjustable to adapt to sensors and reference
- [VADC_convtime] Conversion time below **1 µs** (depending on result width and sample time) [VADC_convtime]

FIR: Finite Impulse Response
IIR: Infinite Impulse Response

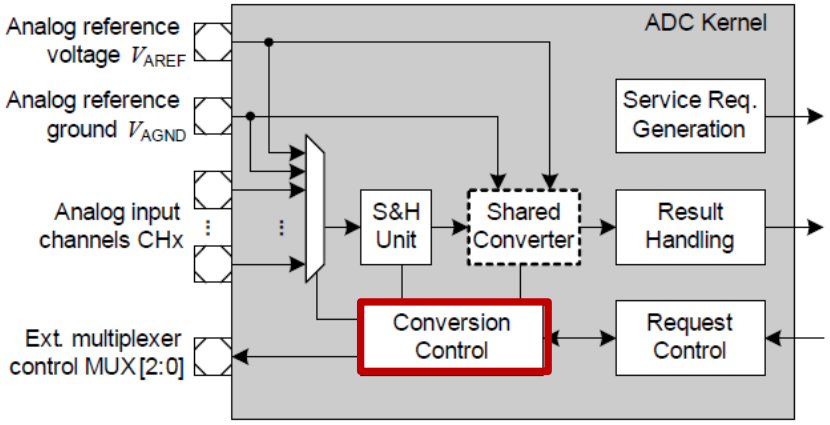
4.1.2 Overview
ADC Kernel and Cluster

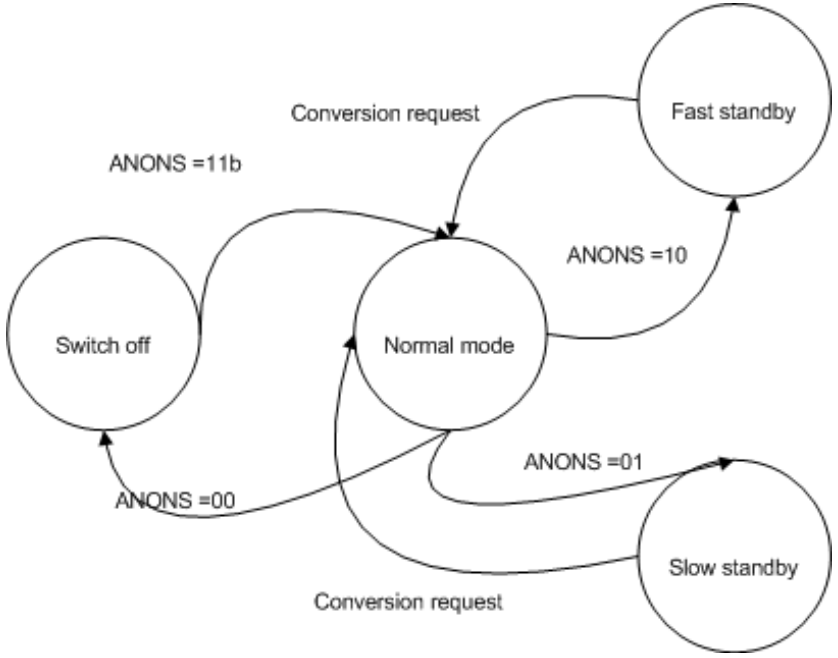
- Kernel : provides a dedicated **Sample&Hold** unit connected to the input multiplexer.
- Cluster : **Several kernels** build–up and share a **common high-speed converter**.



4.1.3 Module Activation and Control
Analog Converter Control

- Configure the operating mode in **GxARBCFG.ANONS**
  - **11b : Normal Operation**
  - 10b : Fast Standby mode
  - 01b : Slow Standby mode
  - 00b : Converter switched Off (default after reset)





4.1.3 Module Activation and Control
Analog Converter Control

- Configure the operating mode in **GxARBCFG.ANONS**

GxARBCFG (x = 0 - 3)
Arbitration Configuration Register, Group x
($x * 0400_H + 0480_H$) **Reset Value: $0000\ 0000_H$**

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SAMPLE	BUSY	CALS	CAL	0	0	SYNRUN	CHNR					CSRC		ANONS	
rh	rh	rh	rh	r	r	rh	rh					rh		rh	

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	0	0	0	ARBM	0	ARBRND		0	0	ANONC	
r	r	r	r	r	r	r	r	rw	r	rw		r	r	rw	

Field	Bits	Type	Description
ANONC	[1:0]	rw	**Analog Converter Control** Defines the value of bitfield ANONS in a stand-alone converter or a converter in master mode. Coding see ANONS or Section 27.4.1.
ANONS	[17:16]	rh	**Analog Converter Control Status** Defined by bitfield ANONC in a stand-alone kernel or a kernel in master mode. In slave mode, this bitfield is defined by bitfield ANONC of the respective master kernel. See also Section 27.4.1. 00_B Analog converter off 01_B Reserved 10_B Reserved 11_B Normal operation (permanently on)

```
Ifx_VADC *adc=&MODULE_VADC;
```

```
adc->G[1].ARBCFG.U=0x3;
    //Normal operation,激活AD转换功能
```

4.1.3 Module Activation and Control
Alternate Reference Selection

SHS0_CALCTR
Calibration Control Register (F002 50BC$_H$) **Reset Value: 0010 0500$_H$**
SHS1_CALCTR
Calibration Control Register (F002 52BC$_H$) **Reset Value: 0010 0500$_H$**

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SU CAL	0	CALMAX						0	SUCALVAL						
w	r	rw						r	rw						

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	CALGNSTC						ARM	AREFDEF			AR CH	ARC DIS	0	CAL ORD
r	r	rw						rw	rw			rw	rw	r	rw

ARCH	3	rw	**Alternate Reference Channel** 0_B Channel 0 1_B Channel 8 *Note: The reference channel must be selected before the calibration is started. Bit ARCH must not be changed during operation.*

AREFDEF	[6:4]	rw	**Alternate Reference Definition** Defines the S&H unit whose alternate reference input (CH0) shall be used if bit ARM = 1. 000_B S&H unit 0 ... 011_B S&H unit 3 Others: Reserved
ARM	7	rw	**Alternate Reference Mode** 0_B Local Mode: Use the alternate reference input (CH0) associated with the corresponding S&H unit 1_B Global Mode: Use the alternate reference input (CH0) of the S&H unit selected by bitfield AREFDEF

- Configure for Alternate selection:
  - ARCH : Alternate Reference Channel
  - ARFEDEF : Alternate Reference Definition
  - ARM : Alternate Reference Mode

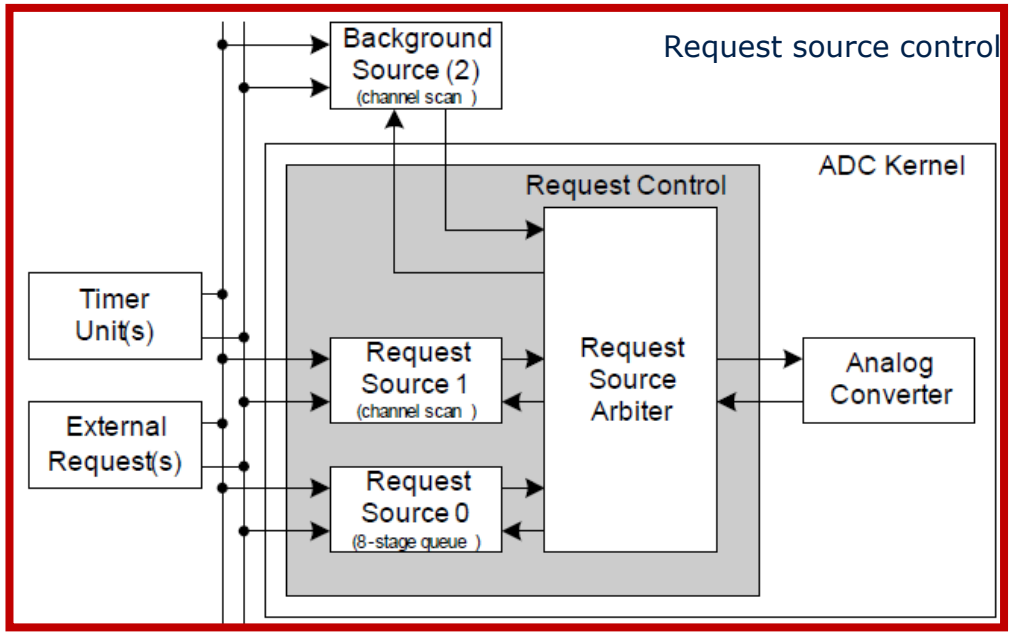
4.1.3 Module Activation and Control
Calibration

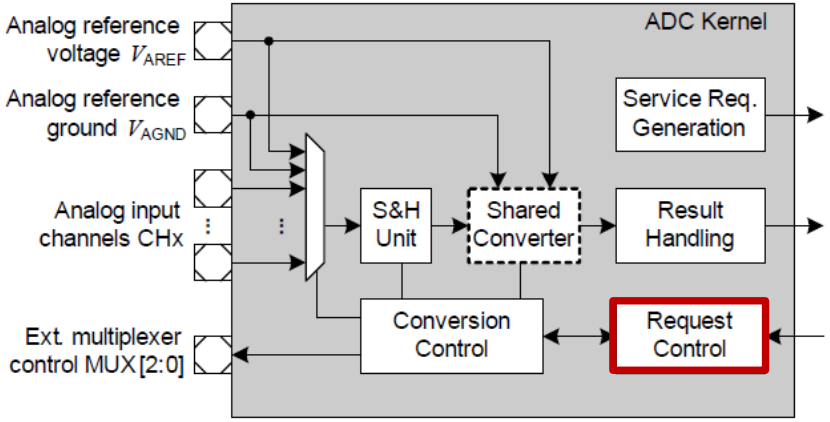
- Calibration automatically compensates deviations caused by
  - Process
  - Temperature
  - Voltage
- Calibration cycles
  - initial start-up calibration (GLOBCFG.SUCAL =1) : 4440 cycles
    - offset calibration: 2(sample) + 8(calibr.) + 3(FSM) = 13 cycles;
    - gain calibration: 5(sample) + 8(calibr.) + 3(FSM) + 12(ref.-switch) = 28 cycles
    - cal. sequence: (1 * offset, 1 * gain std., 1 * gain alt.) per S&H unit:(13 + 28 + 28) * 4 = 276 cycles
    - complete phase: 16 cal. sequences: 16 * 276 = 4 416 cycles = 4 416 * 7.5 ns = **33.12 µs** plus synchronization cycles:

4.1.4 Conversion Request
Overview

- Request:
  - Queued source
  - Scan source
    - Group-specific channel
    - Global channel
- Mode
  - single – slot mode
  - continous mode
- Trigger
  - Software trigger
  - External trigger

通道如何排队？如何触发?





4.1.4 Conversion Request
Queue Request Source

- **A Queued Request Source** (up to 8 channels per group)
  - These channels can be arranged in **an arbitrary （任意的） order**
  - Each entry can be refilled into the buffer to be repeated automatically
  - Start either via software or via a selectable trigger signal from on-chip and external sources

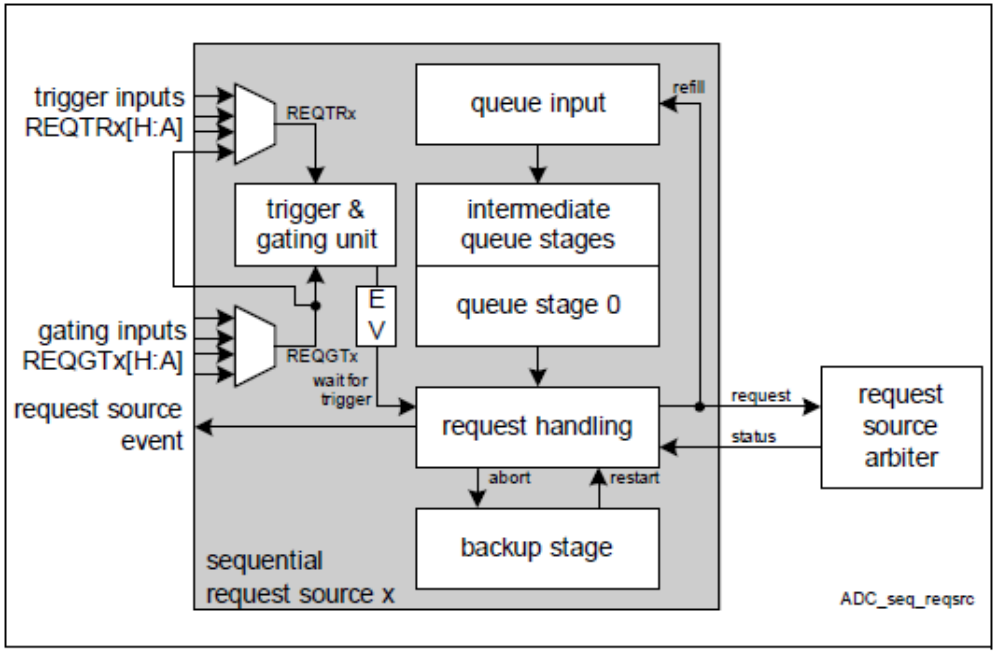


Figure 30-9 Queued Request Source

4.1.4 Conversion Request
Queue Request Source Operation

- Configure the queued request source :
  - Define the sequence by writing the entries to the queue input **GxQINR0** .
  - If hardware trigger or gating is desired,
    - Select the appropriate trigger and gating inputs and the proper transitions by programming **GxQCTRL0**.
    - Enable the trigger and select the gating mode by programming bitfield **ENGT** in register **GxQMR0**.
  - Enable the corresponding arbitration **slot 0** to accept conversion requests from the queued source (see register **GxARBPR**).
- Start a queued sequence :
  - Generate the configured transition at the selected input signal(hardware trigger).
  - Generate a software trigger event by setting **GxQMR0.TREV** = 1.
  - Write a new entry to the queue input of an empty queue.

4.1.4 Conversion Request
Scan Request Source

- 2 scan sources (a linear sequence) :
  - Group scan source : select channel by the select bit in register **GxASSEL**
  - Background scan source : select channel by the select bit in register **BRSSELx**

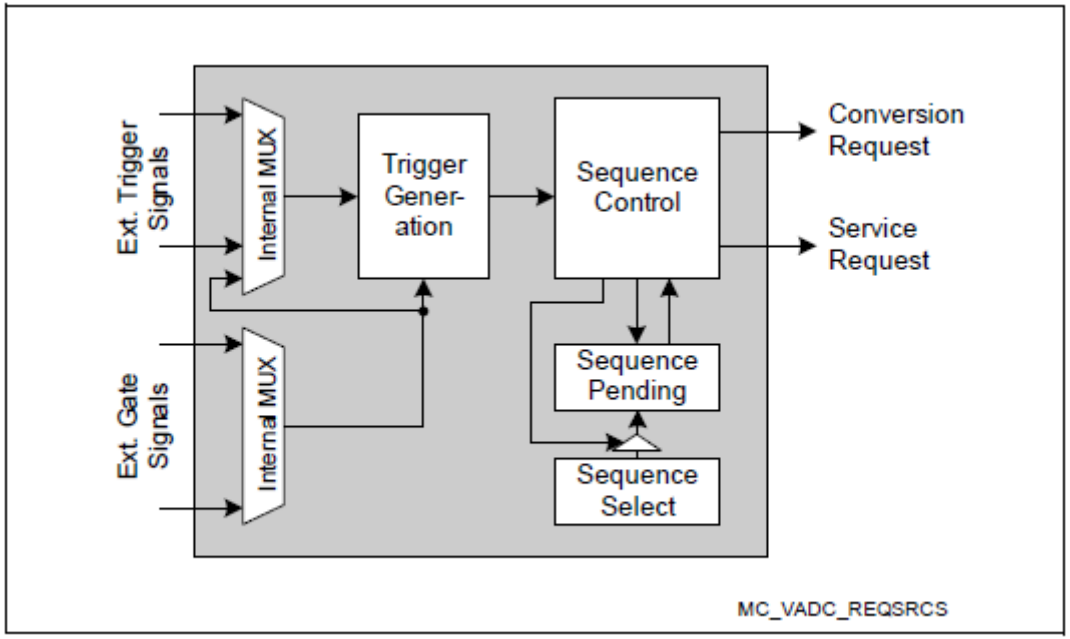


Figure 30-11 Scan Request Source

4.1.4 Conversion Request
Scan Request Source Operation (1)

- Configure the scan request source :
  - Select channels in **GxASSEL or BRSSELx**
  - If hardware trigger or gating is desired,
    - select the appropriate trigger and gating inputs and the proper signal transitions by programming **GxASCTRL** or **BRSCTRL.**
    - Enable the trigger and select the gating mode by programming **GxASMR** or **BRSMR.**
  - Configure the load event operation(**GxASMR** or **BRSMR).**
    - LDM = 0: Overwrite the content of **GxASSEL** or **BRSSELx** to **GxASPND or BRSPNDx**.
    - LDM = 1: OR-combines the content of **GxASSEL** or **BRSSELx to GxASPND or BRSPNDx**.
  - Enable the corresponding arbitration **slot 1** to accept conversion requests from the channel scan source (**GxARBPR**).

4.1.4 Conversion Request
Scan Request Source Operation (2)

- Start a channel scan sequence:
  - Generate the configured transition at the selected input signal (hardware trigger mode) .
  - Generate a software load event by setting LDEV = 1 (**GxASMR** or **BRSMR**).
  - Generate a load event by writing the scan pattern directly to the pending bits in **GxASPND** or **BRSPNDx**.

4.1.4 Conversion Request
Request Source Arbitration Overview

- request source arbiter regularly polls the request sources, one after the other, for pending conversion requests
- each request source is assigned to a certain time slot within an arbitration round, called arbitration slot
  - Arbitration slot 0: **8-stage queued source**, 8-stage sequences in arbitrary order
  - Arbitration slot 1: **n-scan source**, with SAR group input channels (with n=mux width)
  - Arbitration slot 2: **virtual background scan source**, sequences in defined order
  - Last arbitration slot: **Synchronization source**, synchronized conversion requests from another ADC kernel (always handled with the highest priority in a synchronization slave kernel)

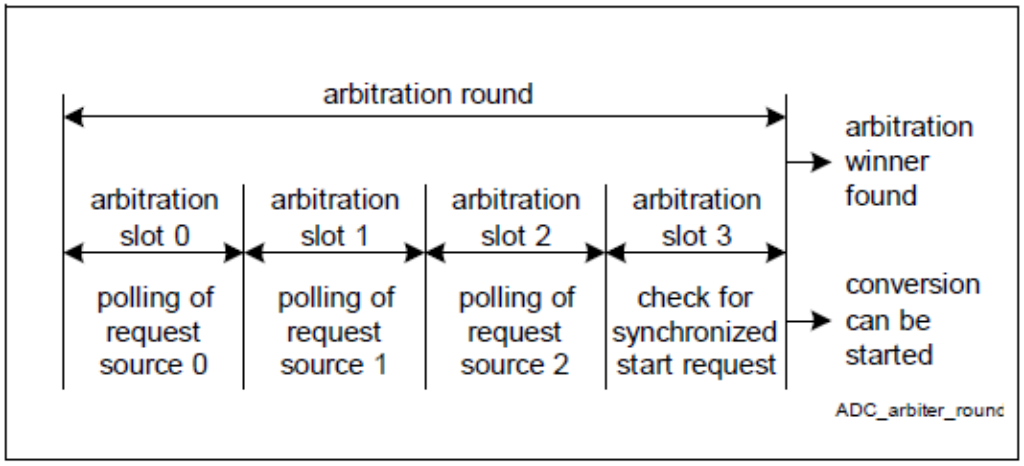


Figure 30-12 Arbitration Round with 4 Arbitration Slots

4.1.4 Conversion Request
Request Source Arbitration：Configuration

GxARBCFG (x = 0 - 7)
Arbitration Configuration Register, Group x
($x * 0400_H + 0480_H$) **Reset Value: $0000\ 0000_H$**

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SAMPLE	BUSY	CALS	CAL	0	0	SYNRUN	CHNR					CSRC		ANONS	
rh	rh	rh	rh	r	r	rh	rh					rh		rh	

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	0	0	0	ARBM	0	ARBRND		0	0	ANONC	
r	r	r	r	r	r	r	r	rw	r	rw		r	r	rw	

Field	Bits	Type	Description
ANONC	[1:0]	rw	**Analog Converter Control** Defines the value of bitfield ANONS in a stand-alone converter or a converter in master mode. Coding see ANONS or **Section 30.4**.
0	[3:2]	r	**Reserved, write 0, read as 0**
ARBRND	[5:4]	rw	**Arbitration Round Length** Defines the number of arbitration slots per arb. round (arbitration round length = t_{ARB}).[1)] 00_B 4 arbitration slots per round ($t_{ARB} = 4 / f_{ADCD}$) 01_B 8 arbitration slots per round ($t_{ARB} = 8 / f_{ADCD}$) 10_B 16 arbitration slots per round ($t_{ARB} = 16 / f_{ADCD}$) 11_B 20 arbitration slots per round ($t_{ARB} = 20 / f_{ADCD}$)
0	6	r	**Reserved, write 0, read as 0**

GxARBPR (x = 0 - 7)
Arbitration Priority Register, Group x
($x * 0400_H + 0484_H$) **Reset Value: $0000\ 0000_H$**

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0	0	0	0	0	ASEN2	ASEN1	ASEN0	0	0	0	0	0	0	0	0
r	r	r	r	r	rw	rw	rw	r	r	r	r	r	r	r	r

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	CSM2	0	PRIO2		CSM1	0	PRIO1		CSM0	0	PRIO0	
r	r	r	r	rw	r	rw		rw	r	rw		rw	r	rw	

ASENy **(y = 0 - 2)**	24 + y	rw	**Arbitration Slot y Enable** Enables the associated arbitration slot of an arbiter round. The request source bits are not modified by write actions to ASENR. 0_B The corresponding arbitration slot is disabled and considered as empty. Pending conversion requests from the associated request source are disregarded. 1_B The corresponding arbitration slot is enabled. Pending conversion requests from the associated request source are arbitrated.

- Duration of an arbitration slot：tSlot = (DIVD+1) / fADC **(DIVD-仲裁时钟的分频系数)**
- Duration of an arbitration round, tARB = 4 × tSlot

4.1.4 Conversion Request
Request Source Arbitration：Start Modes

- Converter is currently **IDLE**
  - Start the winner immediately
- Current conversion has **same or higher priority**
  - Waiting for the completion of the current conversion
- Current conversion has **lower priority** - the action is user-configurable
  - **Wait-for-start mode 优哉模式**
    - Start the new conversion after the current conversion is completed.
      - Provides maximum throughput, but can produce a jitter for the higher priority conversion
  - **Cancel-inject-repeat mode 十万火急模式**
    - the current conversion is aborted, then start the new conversion of the arbitration winner (3 *f*ADC cycles)
    - The aborted conversion request is restored in the next arbitration round
      - Provides minimum jitter for the higher priority conversions, but reduces the overall throughput
    - Exception
      - if both requests target the same result register with wait-for-read mode active, the current conversion cannot be aborted

4.1.4 Conversion Request
Request Source Arbitration : Start Modes

- Conversion start mode can be individually programmed for each request source by bits in SFR **GxARBCFG** and is applied to all channels requested by the source
  - In the example below, channel A is issued by a request source with a lower priority than the request source requesting the conversion of channel B

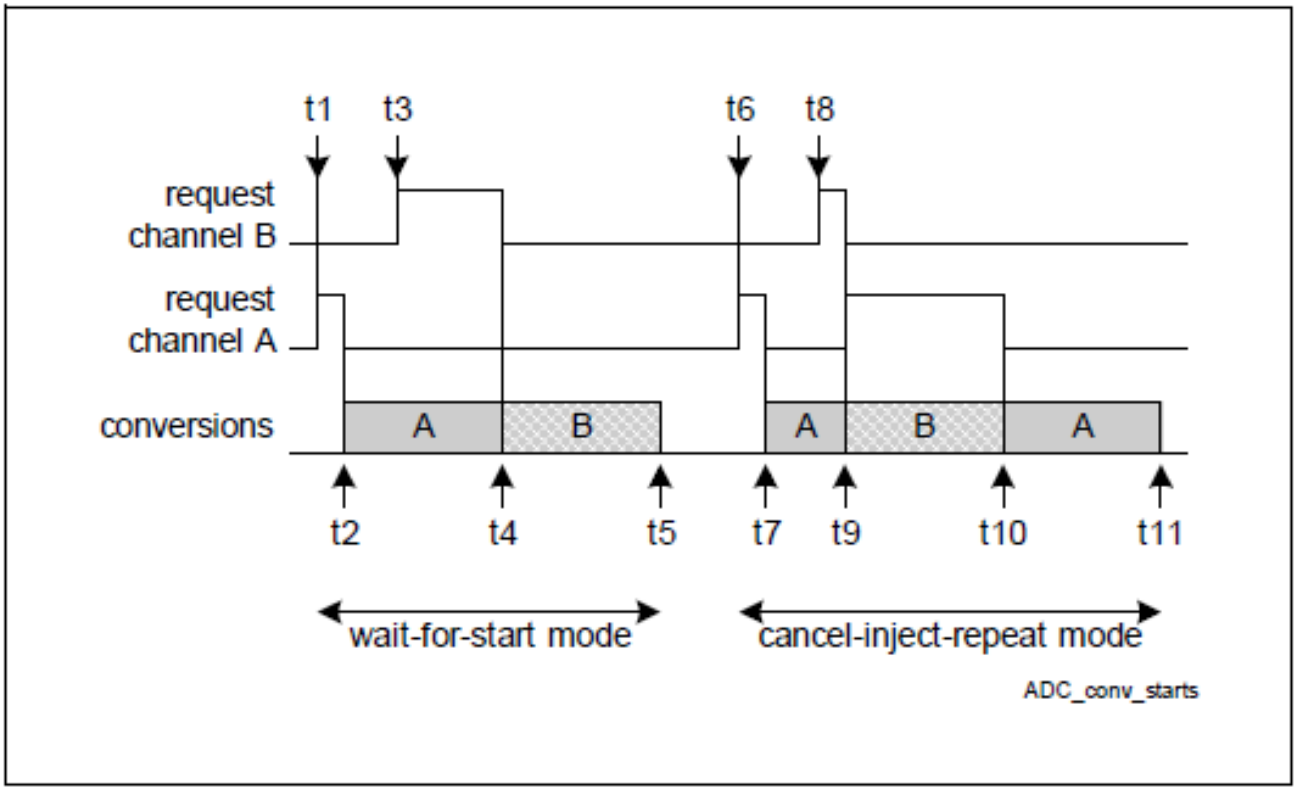
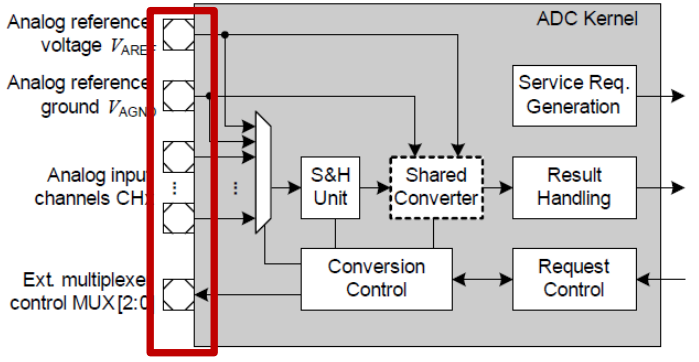


Figure 30-13 Conversion Start Modes

4.1.5 Input Channel Configuration
Overview

Configured parameters :

- **Channel Parameters**
  - sample time & data width of the result : (defined in input classes)
- **Reference selection**
  - Standard reference (VAREF) or Alternate reference (CH0)
- **Result target**
  - group-specific result register : channel-specific or source-specific
  - global result register
- **Result position**
  - result values can be stored left-aligned or right-aligned
- **Broke Wire Detection**
- **Synchronization of Conversion**
- **Alias Feature**
  - conversion requests for channels CH0 and/or CH1 can be redirected to other channels
- **Channel event handling**
  - channel events can be generated whenever a new result value becomes available
  - channel event generation can be restricted to values that lie inside or outside a user-configurable band (**limit check**)



4.1.5 Input Channel Configuration
Channel parameters

- Channel control registers **GxCHCTRy**

G0CHCTRy (y = 0 - 11)
Group 0, Channel y Ctrl. Reg. (0600_H + y * 0004_H) Reset Value: 0000 0000_H
G1CHCTRy (y = 0 - 11)
Group 1, Channel y Ctrl. Reg. ($0A00_H$ + y * 0004_H) Reset Value: 0000 0000_H
G2CHCTR0
Group 2, Channel 0 Ctrl. Reg. ($0E00_H$) Reset Value: 0000 0000_H
G3CHCTR0
Group 3, Channel 0 Ctrl. Reg. (1200_H) Reset Value: 0000 0000_H

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0	BWD EN	BWD CH		0	0	0	0	0	0	RES POS	RES TBS	RESREG			
r	rw	rw		r	r	r	r	r	r	rw	rw	rw			

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BNDSELX				REF SEL	SY NC	CHEV MODE		BNDSELU		BNDSELL		0	0	ICLSEL	
rw				rw	rw	rw		rw		rw		r	r	rw	

Field	Bits	Type	Description
ICLSEL	[1:0]	rw	**Input Class Select** 00_B Use group-specific class 0 01_B Use group-specific class 1 10_B Use global class 0 11_B Use global class 1

Input Class Registers

The group-specific input class registers define the sample time and data conversion mode for each channel of the respective group that selects them via bitfield ICLSEL in its channel control register GxCHCTRx.

The global input class registers define the sample time and data conversion mode for each channel of any group that selects them via bitfield ICLSEL in its channel control register GxCHCTRx.

4.1.5 Input Channel Configuration
Channel parameters

Input Class Registers

GxICLASS0 (x = 0 - 3)
Input Class Register 0, Group x
($x * 0400_H + 04A0_H$) **Reset Value: $0000\ 0000_H$**

GxICLASS1 (x = 0 - 3)
Input Class Register 1, Group x
($x * 0400_H + 04A4_H$) **Reset Value: $0000\ 0000_H$**

GLOBICLASSy (y = 0 - 1)
Input Class Register y, Global
($00A0_H + y * 0004_H$) **Reset Value: $0000\ 0000_H$**

Field	Bits	Type	Description
STCS	[4:0]	rw	**Sample Time Control for Standard Conversions** Number of additional clock cycles to be added to the minimum sample phase of 2 analog clock cycles: Coding and resulting sample time see Table 27-4. For conversions of external channels, the value from bitfield STCE can be used.
0	[7:5]	r	**Reserved, write 0, read as 0**
CMS	[10:8]	rw	**Conversion Mode for Standard Conversions** 000_B 12-bit conversion 001_B 10-bit conversion 010_B 8-bit conversion 011_B Reserved 100_B Reserved 101_B 10-bit fast compare mode 110_B Reserved 111_B Reserved
0	[15:11]	r	**Reserved, write 0, read as 0**

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0	0	0	0	0	CME			0	0	0	STCE				
r	r	r	r	r	rw			r	r	r	rw				
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	CMS			0	0	0	STCS				
r	r	r	r	r	rw			r	r	r	rw				

Table 30-5 Sample Time Coding

STCS / STCE	Additional Clock Cycles	Sample Time
$0\ 0000_B$	0	$2 / f_{ADCI}$
$0\ 0001_B$	1	$3 / f_{ADCI}$
...	...	...
$0\ 1111_B$	15	$17 / f_{ADCI}$
$1\ 0000_B$	16	$18 / f_{ADCI}$
$1\ 0001_B$	32	$34 / f_{ADCI}$
...	...	...
$1\ 1110_B$	240	$242 / f_{ADCI}$
$1\ 1111_B$	256	$258 / f_{ADCI}$

4.1.5 Input Channel Configuration
Conversion Modes

- A conversion can be executed in several ways, bitfield **CMS/CME** in register **GxICLASS0 (x = 0 - 3)** etc. to select a mode.
  - Standard Conversions
  - Fast Compare Mode : the selected input voltage is directly compared with a digital value that is stored in the corresponding result register

Fast Compare Mode

In Fast Compare Mode, the selected input voltage is directly compared with a digital value that is stored in the corresponding result register. This compare operation returns a binary result indicating if the compared input voltage is above or below the given reference value. This result is generated quickly and thus supports monitoring of boundary values.

Fast Compare Mode uses a 10-bit compare value stored left-aligned at bit position 11. Separate positive and negative delta values define an arbitrary hysteresis band.

4.1.5 Input Channel Configuration
Alias Feature

- Alias Feature redirects conversion requests for channels CH0 and/or CH1 to other channel numbers

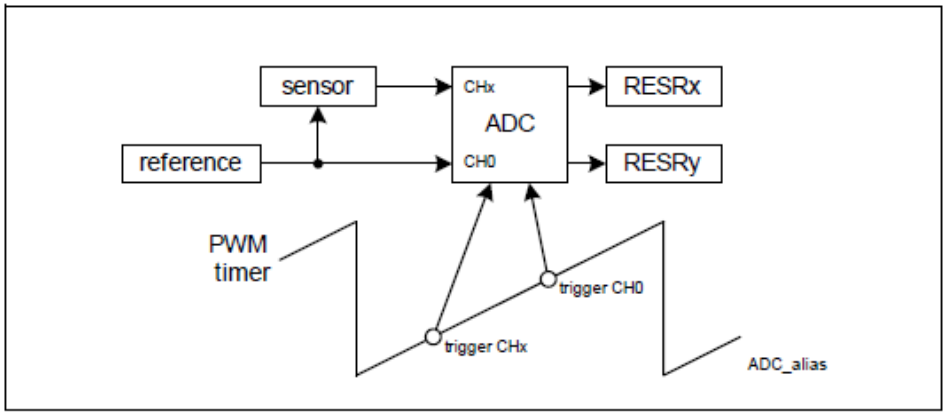


Figure 30-14 Alias Feature

Field	Bits	Type	Description
ALIAS0	[4:0]	rw	**Alias Value for CH0 Conversion Requests** Indicates the channel that is converted instead of channel CH0. The conversion is done with the settings defined for channel CH0.
0	[7:5]	r	**Reserved, write 0, read as 0**
ALIAS1	[12:8]	rw	**Alias Value for CH1 Conversion Requests** Indicates the channel that is converted instead of channel CH1. The conversion is done with the settings defined for channel CH1.
0	[31:13]	r	**Reserved, write 0, read as 0**

GxALIAS (x = 0 - 7)
Alias Register, Group x **($x * 0400_H + 04B0_H$)** **Reset Value: $0000\ 0100_H$**

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	ALIAS1					0	0	0	ALIAS0				
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	rw					r	r	r	rw				

The Alias Feature redirects conversion requests for channels CH0 and/or CH1 to other channel numbers. This feature can be used to trigger conversions of the same input channel by independent events and to store the conversion results in different result registers.

4.1.5 Input Channel Configuration
Limit Checking and Channel Event Generation

- Limit checking mechanism automatically compares each conversion result to an upper and a lower boundary value.
  - Bitfields **BNDSELL, BNDSELU, BNDSELX** in the corresponding channel control register **GxCHCTRy** select the valid boundary value
- A **channel event** can then be generated when the result of a conversion/comparison is inside or outside a user-defined band

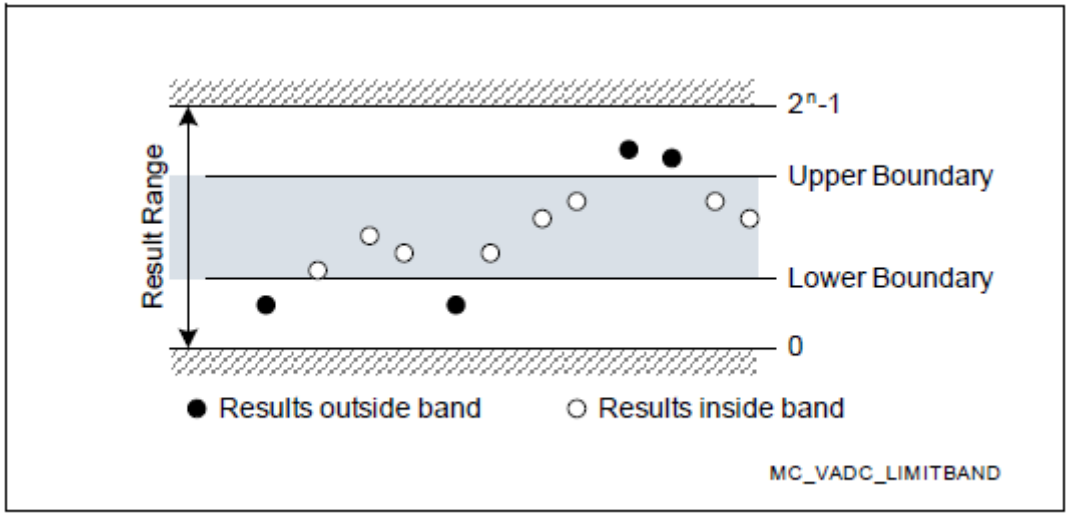


Figure 30-15 Result Monitoring through Limit Checking

BNDSELL	[5:4]	rw	**Lower Boundary Select**[1)] 00_B Use group-specific boundary 0 01_B Use group-specific boundary 1 10_B Use global boundary 0 11_B Use global boundary 1
BNDSELU	[7:6]	rw	**Upper Boundary Select**[1)] 00_B Use group-specific boundary 0 01_B Use group-specific boundary 1 10_B Use global boundary 0 11_B Use global boundary 1

BNDSELX	[15:12]	rw	**BoundaryExtension**[1)] 0000_BStandard mode: select boundaries via BNDSELU/BNDSELL 0001_BUse result reg. GxRES1 as upper boundary … 1111_BUse result reg. GxRES15 as upper boundary

4.1.5 Input Channel Configuration
GxCHCTRy, GxBOUND and GLOBBOUND

G0CHCTRy (y = 0 - 11)
Group 0, Channel y Ctrl. Reg. ($0600_H + y * 0004_H$) **Reset Value: 0000 0000$_H$**
G1CHCTRy (y = 0 - 11)
Group 1, Channel y Ctrl. Reg. ($0A00_H + y * 0004_H$) **Reset Value: 0000 0000$_H$**
G2CHCTR0
Group 2, Channel 0 Ctrl. Reg. ($0E00_H$) **Reset Value: 0000 0000$_H$**
G3CHCTR0
Group 3, Channel 0 Ctrl. Reg. (1200_H) **Reset Value: 0000 0000$_H$**

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0	BWD EN	BWD CH		0	0	0	0	0	0	RES POS	RES TBS	RESREG			
r	rw	rw		r	r	r	r	r	r	rw	rw	rw			

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BNDSELX				REF SEL	SY NC	CHEV MODE		BNDSELU		BNDSELL		0	0	ICLSEL	
rw				rw	rw	rw		rw		rw		r	r	rw	

Field	Bits	Type	Description
BNDSELL	[5:4]	rw	**Lower Boundary Select**[1)] 00_B Use group-specific boundary 0 01_B Use group-specific boundary 1 10_B Use global boundary 0 11_B Use global boundary 1
BNDSELU	[7:6]	rw	**Upper Boundary Select**[1)] 00_B Use group-specific boundary 0 01_B Use group-specific boundary 1 10_B Use global boundary 0 11_B Use global boundary 1

GxBOUND (x = 0 - 3)
Boundary Select Register, Group x
($x * 0400_H + 04B8_H$) **Reset Value: 0000 0000$_H$**
GLOBBOUND
Global Boundary Select Register ($00B8_H$) **Reset Value: 0000 0000$_H$**

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	BOUNDARY1												0	0	0	0	BOUNDARY0											
r	r	r	r	rw												r	r	r	r	rw											

Field	Bits	Type	Description
BOUNDARY0	[11:0]	rw	**Boundary Value 0 for Limit Checking** Standard Mode: This value is compared against the left-aligned conversion result. Fast Compare Mode: This value is added to the reference value (upper delta).
0	[15:12]	r	**Reserved, write 0, read as 0**
BOUNDARY1	[27:16]	rw	**Boundary Value 1 for Limit Checking** Standard Mode: This value is compared against the left-aligned conversion result. Fast Compare Mode: This value is subtracted from the reference value (lower delta).
0	[31:28]	r	**Reserved, write 0, read as 0**

4.1.5 Input Channel Configuration
Fast Compare Mode

- The input signal is directly compared to a value stored in bitfield RESULT in the associated result register
- A **channel event** can then be generated when the input signal becomes higher (or lower) than the compare value

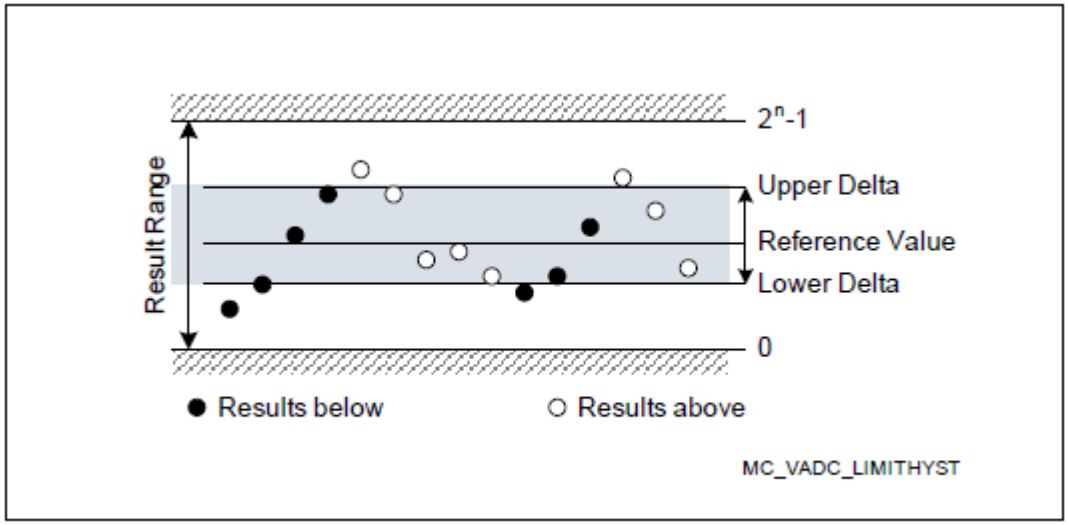


Figure 30-16 Result Monitoring through Compare with Hysteresis

CMS	[10:8]	rw	**Conversion Mode for Standard Conversions** 000_B 12-bit conversion 001_B 10-bit conversion 010_B 8-bit conversion 011_B Reserved 100_B Reserved 101_B 10-bit fast compare mode 110_B Reserved 111_B Reserved

GxICLASS0 (x = 0 - 3)
Input Class Register 0, Group x
($x * 0400_H + 04A0_H$) **Reset Value: $0000\ 0000_H$**

GxICLASS1 (x = 0 - 3)
Input Class Register 1, Group x
($x * 0400_H + 04A4_H$) **Reset Value: $0000\ 0000_H$**

GLOBICLASSy (y = 0 - 1)
Input Class Register y, Global
($00A0_H + y * 0004_H$) **Reset Value: $0000\ 0000_H$**

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0	0	0	0	0	CME			0	0	0	STCE				
r	r	r	r	r	rw			r	r	r	rw				

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	CMS			0	0	0	STCS				
r	r	r	r	r	rw			r	r	r	rw				

4.1.6 Conversion Timing
Overview

- The total time required for a conversion comprises the time from the **start** of the sample phase until the **availability** of the result. Two modes can be selected:
  - **Compatible timing mode**.
  - **Accelerated timing mode**.

4.1.6 Conversion Timing
Compatible Time Mode

- The conversion timing depends on the following user-definable factors:
  - The ADC conversion clock **frequency**, where $f_{ADCI} = f_{ADC} / (\mathbf{DIVA}+1)$ (the minimum prescaler factor for AURIX = 2)
  - The selected **sample time**, where $t_S = (2 + STC) \times t_{ADCI}$ (STC = additional sample time)
  - The **result width** N (8/10/12 bits) for normal conversions
  - The **post-calibration** time PC, if selected (PC = 2), otherwise 0
  - **Synchronization** steps done at module clock speed
- The conversion time is the sum of sample time, conversion steps, and synchronization.
  - $t_{CN} = (2 + \mathbf{STC} + N + PC) \times t_{ADCI} + 2 \times t_{ADC}$ **理想情况认为STC=0**

Timing Examples

System assumptions:
f_{ADC} = 100 MHz i.e. t_{ADC} = 10 ns, DIVA = 5, f_{ADCI} = 16.67 MHz i.e. t_{ADCI} = 60 ns

According to the given formulas the following minimum conversion times can be achieved:

12-bit calibrated conversion:
$t_{CN12C} = (2 + 12 + 2) \times t_{ADCI} + 2 \times t_{ADC} = 16 \times 60\ ns + 2 \times 10\ ns = 980\ ns$

10-bit uncalibrated conversion:
$t_{CN10} = (2 + 10) \times t_{ADCI} + 2 \times t_{ADC} = 12 \times 60\ ns + 2 \times 10\ ns = 740\ ns$

4.1.6 Conversion Timing
Accelerated Time Mode

- The total time required for a conversion depends on several user-definable factors:
  - The ADC conversion clock **frequency**, where $f_{ADCI} = f_{ADC} / (DIVA+1)$ (the minimum prescaler factor for AURIX = 2)
  - The selected sample time, where $t_S = (2 + STC) \times t_{ADCI}$ (STC = additional sample time)
  - The result width N (8/10/12 bits) for normal conversions
  - The post-calibration time PC, if selected (PC = 2, otherwise 0)
  - The position of the conversion within the stepper sequence
  - Synchronization steps done at module clock speed
- The conversion time is the sum of sample time, conversion steps, and synchronization.
  - t_{CN} = (SST +5) x tVADC + (N+10) x tSH +(4 x tSH)

System assumptions:
f_{ADC} = 133.33 MHz, DIVS = 0, i.e. $t_{SH} = t_{ADC}$ = 7.5 ns,
f_{VADC} = 100 MHz, i.e. t_{VADC} = 10 ns,
SST = 5, i.e. t_S = 50 ns

According to the given formula the following minimum conversion times can be achieved:
(The numbers in parentheses apply for the 2nd, 3rd, 4th conversion of a round)

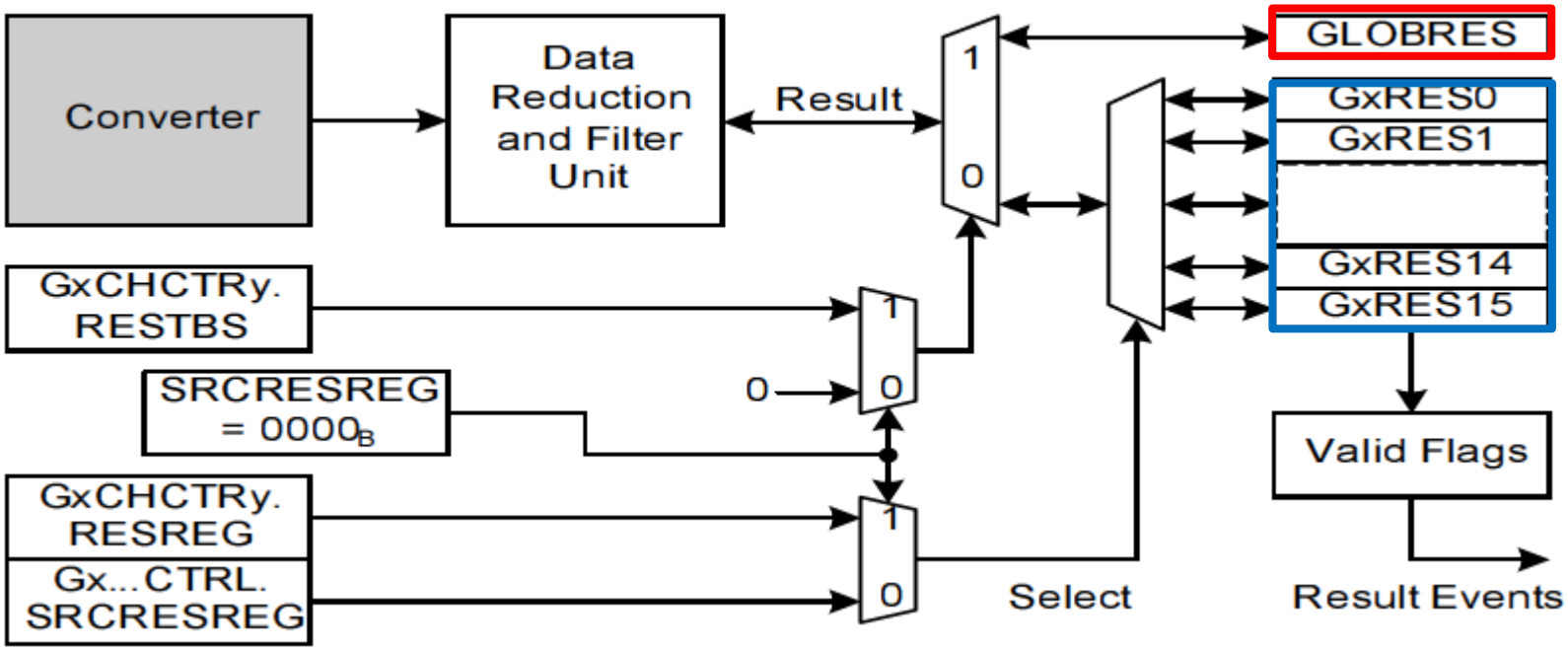
12-bit conversion:
$t_{CN12} = 10 \times t_{VADC} + 26 \times t_{SH}$ = 100 ns + 26 × 7.5 ns = 295 ns (460 ns, 625 ns, 790 ns)

4.1.7 Conversion Result Handling
Overview

- Conversion result handling comprises the following functions :
  - **Storage of Conversion Results**
  - **Data Alignment**
  - **Wait-for-Read Mode**
  - **Result Event Generation**
  - **Data reduction or anti-aliasing filtering**

4.1.7 Conversion Result Handling
Result storage

- The conversion result values of a certain group can be stored in :
  - **Group Result Register :** returns the result value and the channel number
  - **Global Result Register :** returns the result value and the channel number and the group number



4.1.7 Conversion Result Handling
Data Alignment

- The position of a conversion result value within the selected result register depends on 3 configurations
  - selected result width (12/10/8 bits)
  - selected result position (Left/Right-aligned)
  - selected data accumulation mode (data reduction)

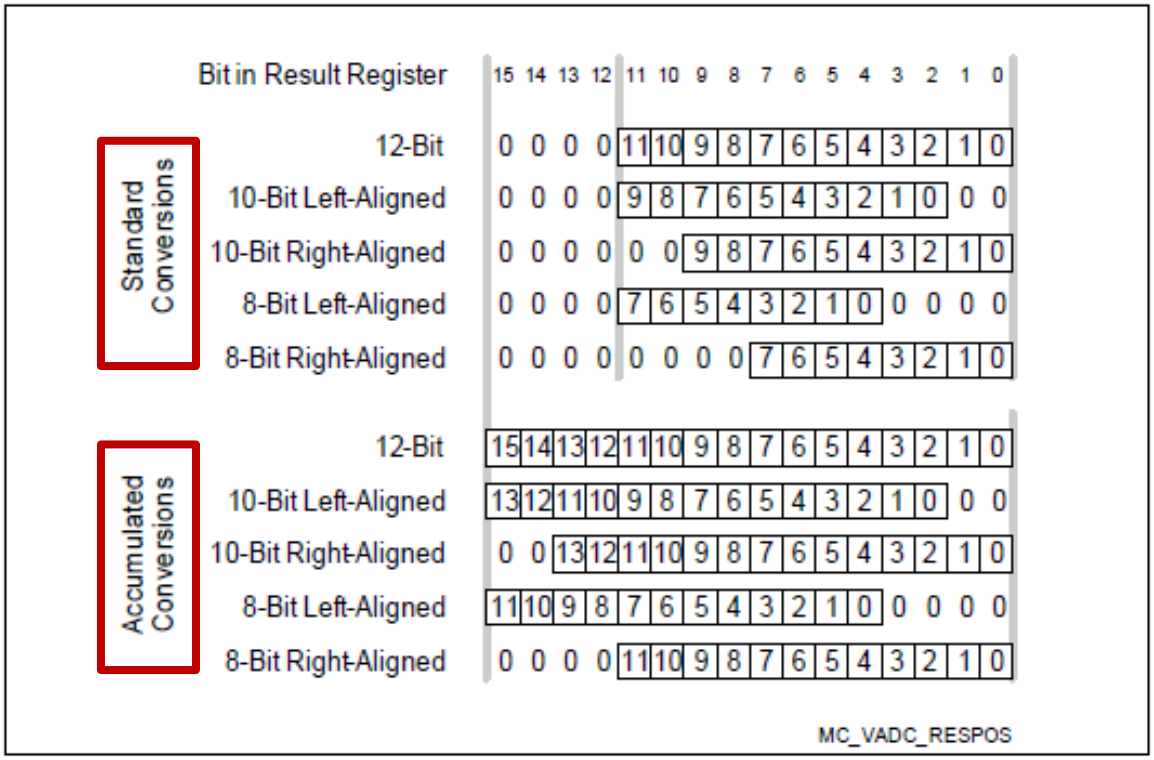
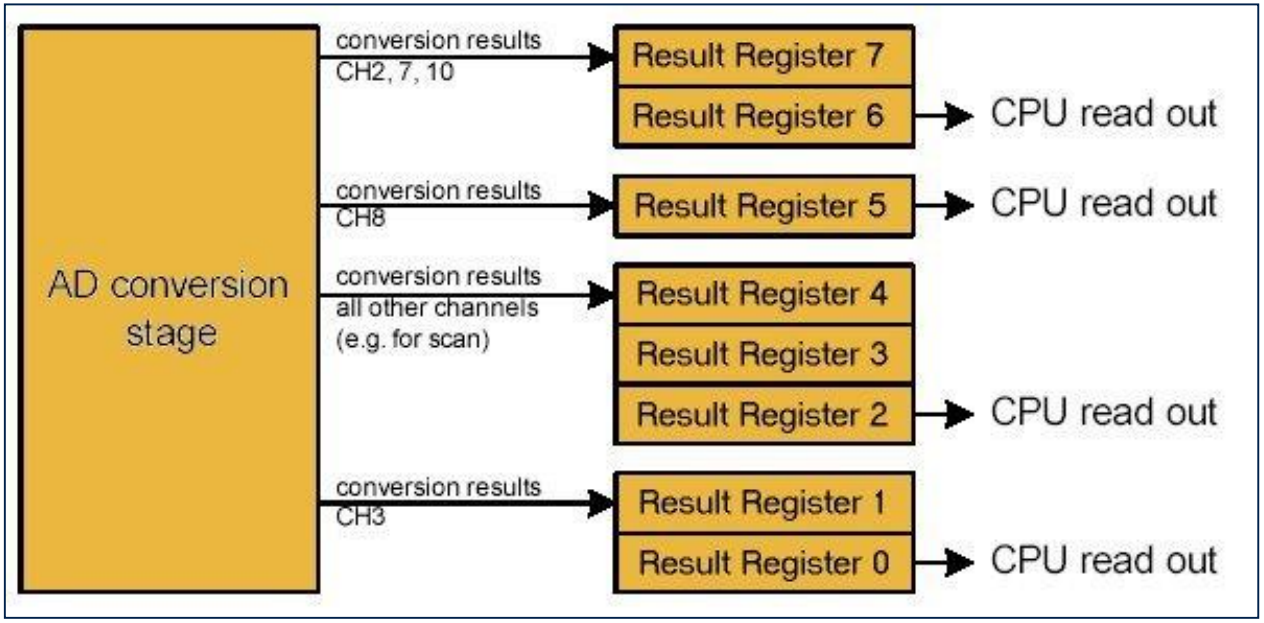


Figure 30-20 Result Storage Options

4.1.7 Conversion Result Handling
Result FIFO

- Several **consecutive（连续的）** result registers can be organized in FIFO (First In First Out **先进先出**) structure (up to 16 stages).
- This functionality aims at relaxing latency constraints without data loss. It can be combined with data reduction filters.
- An **interrupt** can be generated when the **Valid Flag** of the last result register is set.

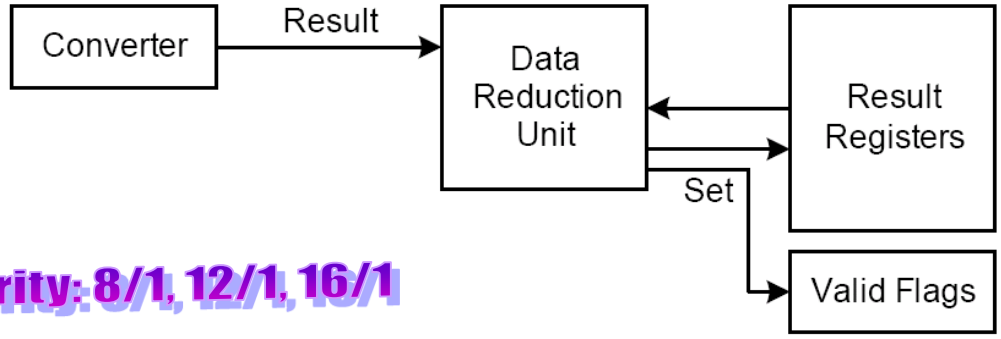


4.1.7 Conversion Result Handling
Wait-for-Read Mode & Result Event Generation

- Wait-for-Read Mode :
  - Prevents data loss due to overwriting a result register with a new conversion result.
- Result Event Generation:
  - A result event can be generated when a new value is stored to a result register.

4.1.7 Conversion Result Handling
Application & debug view

- Each **result register** has an individual data valid flag (**VF**) associated with it
  This flag indicates when "new" valid data has been stored in the corresponding result register and can be read out
- Result registers can be read via two different views (using different addresses)
  - When a result register is read via the **application view**, the corresponding valid flag is automatically cleared
    This provides an easy handshake between result generation and retrieval and furthermore also supports wait-for-read mode
  - When a result register is read via the **debug view**, the corresponding valid flag remains unchanged when the result is read
    This supports debugging by delivering the result value without disturbing the handshake with the application



available mux granularity: 8/1, 12/1, 16/1

4.1.7 Conversion Result Handling
Data Modification Overview

- **Data reduction** automatically accumulates a series of conversion results before generating a result interrupt
  - **Standard Data Reduction Mode (for GxRES0 … GxRES15)**
    - Accumulates 2, 3, or 4 result values within each result register
    - can be combined with FIFO mode
  - **FIR Mode(FIR, for GxRES7, GxRES15)**
    - applies a **3rd order** Finite Impulse Response Filter (FIR) with selectable coefficients to the conversion results for the selected result register
  - **IIR Mode(IIR, for GxRES7, GxRES15)**
    - applies a **1st order** Infinite Impulse Response Filter (IIR) with selectable coefficients to the conversion results for the selected result register
  - **Difference Mode (for GxRES1 … GxRES15):**
    - Subtracts the contents of result register GxRES0 from the conversion results for the selected result register. Bitfield DRCTR is not used in this mode.

4.1.7 Conversion Result Handling
Data Modification Overview

- **Data reduction** automatically accumulates a series of conversion results before generating a result interrupt

G0RCRy (y = 0 - 15)
Group 0 Result Control Reg. y ($0680_H + y * 0004_H$) **Reset Value: $0000\ 0000_H$**
G1RCRy (y = 0 - 15)
Group 1 Result Control Reg. y ($0A80_H + y * 0004_H$) **Reset Value: $0000\ 0000_H$**
G2RCRy (y = 0 - 15)
Group 2 Result Control Reg. y ($0E80_H + y * 0004_H$) **Reset Value: $0000\ 0000_H$**
G3RCRy (y = 0 - 15)
Group 3 Result Control Reg. y ($1280_H + y * 0004_H$) **Reset Value: $0000\ 0000_H$**

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SRG EN	0	0	0	0	FEN		WFR	0	0	DMM		DRCTR			
rw	r	r	r	r	rw		rw	r	r	rw		rw			

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r

Field	Bits	Type	Description
0	[15:0]	r	**Reserved, write 0, read as 0**
DRCTR	[19:16]	rw	**Data Reduction Control** Defines how result values are stored/accumulated in this register for the final result. The data reduction counter DRC can be loaded from this bitfield. The function of bitfield DRCTR is determined by bitfield DMM.
DMM	[21:20]	rw	**Data Modification Mode** 00_B Standard data reduction (accumulation) 01_B Result filtering mode[1)] 10_B Difference mode 11_B Reserved See "Data Modification" on Page 27-125

4.1.7 Conversion Result Handling
Data Modification Overview

Table 27-6 Function of Bitfield DRCTR

DRCTR	Standard Data Reduction Mode (DMM = 00_B)	DRCTR	Result Filtering Mode (DMM = 01_B)[1)]
0000_B	Data Reduction disabled	0000_B	FIR filter: a=2, b=1, c=0
0001_B	Accumulate 2 result values	0001_B	FIR filter: a=1, b=2, c=0
0010_B	Accumulate 3 result values	0010_B	FIR filter: a=2, b=0, c=1
0011_B	Accumulate 4 result values	0011_B	FIR filter: a=1, b=1, c=1
0100_B	Reserved	0100_B	FIR filter: a=1, b=0, c=2
0101_B	Reserved	0101_B	FIR filter: a=3, b=1, c=0
0110_B	Reserved	0110_B	FIR filter: a=2, b=2, c=0
0111_B	Reserved	0111_B	FIR filter: a=1, b=3, c=0
1000_B	Reserved	1000_B	FIR filter: a=3, b=0, c=1
1001_B	Reserved	1001_B	FIR filter: a=2, b=1, c=1
1010_B	Reserved	1010_B	FIR filter: a=1, b=2, c=1
1011_B	Reserved	1011_B	FIR filter: a=2, b=0, c=2
1100_B	Reserved	1100_B	FIR filter: a=1, b=1, c=2
1101_B	Reserved	1101_B	FIR filter: a=1, b=0, c=3

Table 27-6 Function of Bitfield DRCTR (cont'd)

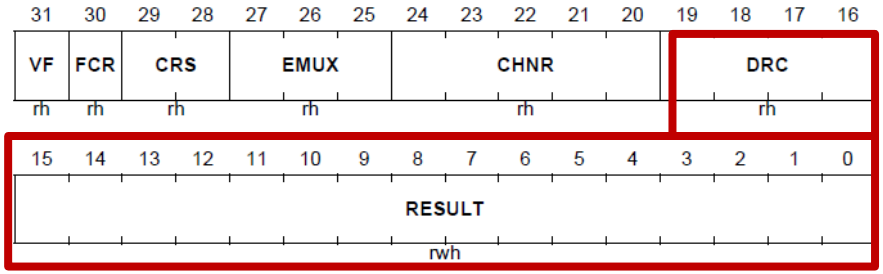
DRCTR	Standard Data Reduction Mode (DMM = 00_B)	DRCTR	Result Filtering Mode (DMM = 01_B)[1)]
1110_B	Reserved	1110_B	IIR filter: a=2, b=2
1111_B	Reserved	1111_B	IIR filter: a=3, b=4

1) The filter registers are cleared while bitfield DMM ≠ 01_B.

4.1.7 Conversion Result Handling
Data Modification Overview

- **Data reduction** automatically accumulates a series of conversion results before generating a result interrupt
- **DRC-Data Reduction Counter**

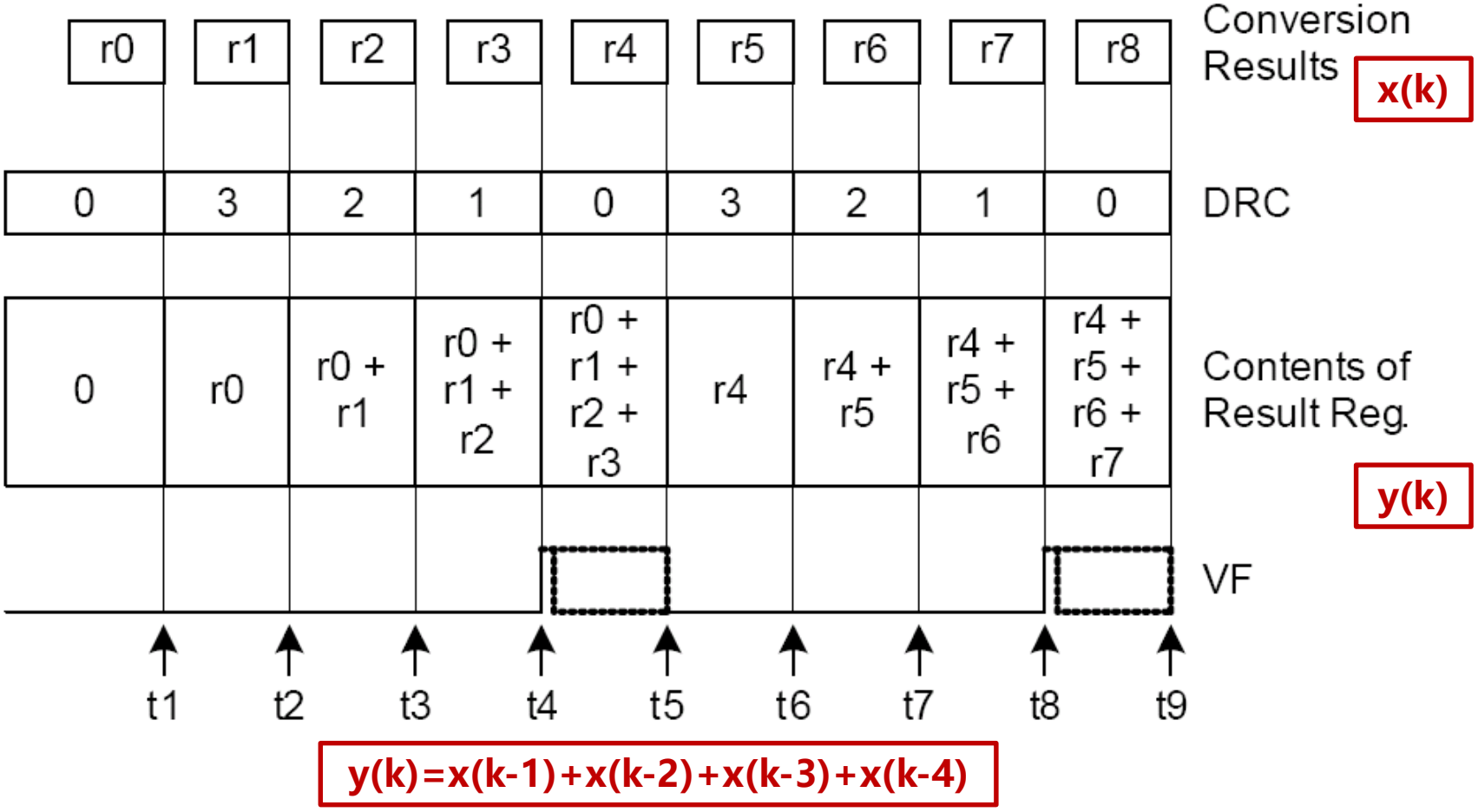
G0RESy (y = 0 - 15)
Group 0 Result Register y ($0700_H + y * 0004_H$) Reset Value: $0000\ 0000_H$
G1RESy (y = 0 - 15)
Group 1 Result Register y ($0B00_H + y * 0004_H$) Reset Value: $0000\ 0000_H$
G2RESy (y = 0 - 15)
Group 2 Result Register y ($0F00_H + y * 0004_H$) Reset Value: $0000\ 0000_H$
G3RESy (y = 0 - 15)
Group 3 Result Register y ($1300_H + y * 0004_H$) Reset Value: $0000\ 0000_H$



Field	Bits	Type	Description
RESULT	[15:0]	rwh	**Result of Most Recent Conversion** The position of the result bits within this bitfield depends on the configured operating mode. Please, refer to Section 27.9.2.
DRC	[19:16]	rh	**Data Reduction Counter** Indicates the number of values still to be accumulated for the final result. The final result is available and valid flag VF is set when bitfield DRC becomes zero (by decrementing or by reload). See "Data Modification" on Page 27-125
CHNR	[24:20]	rh	**Channel Number** Indicates the channel number corresponding to the value in bitfield RESULT.

4.1.7 Conversion Result Handling
Data Modification : Standard Data Reduction Mode

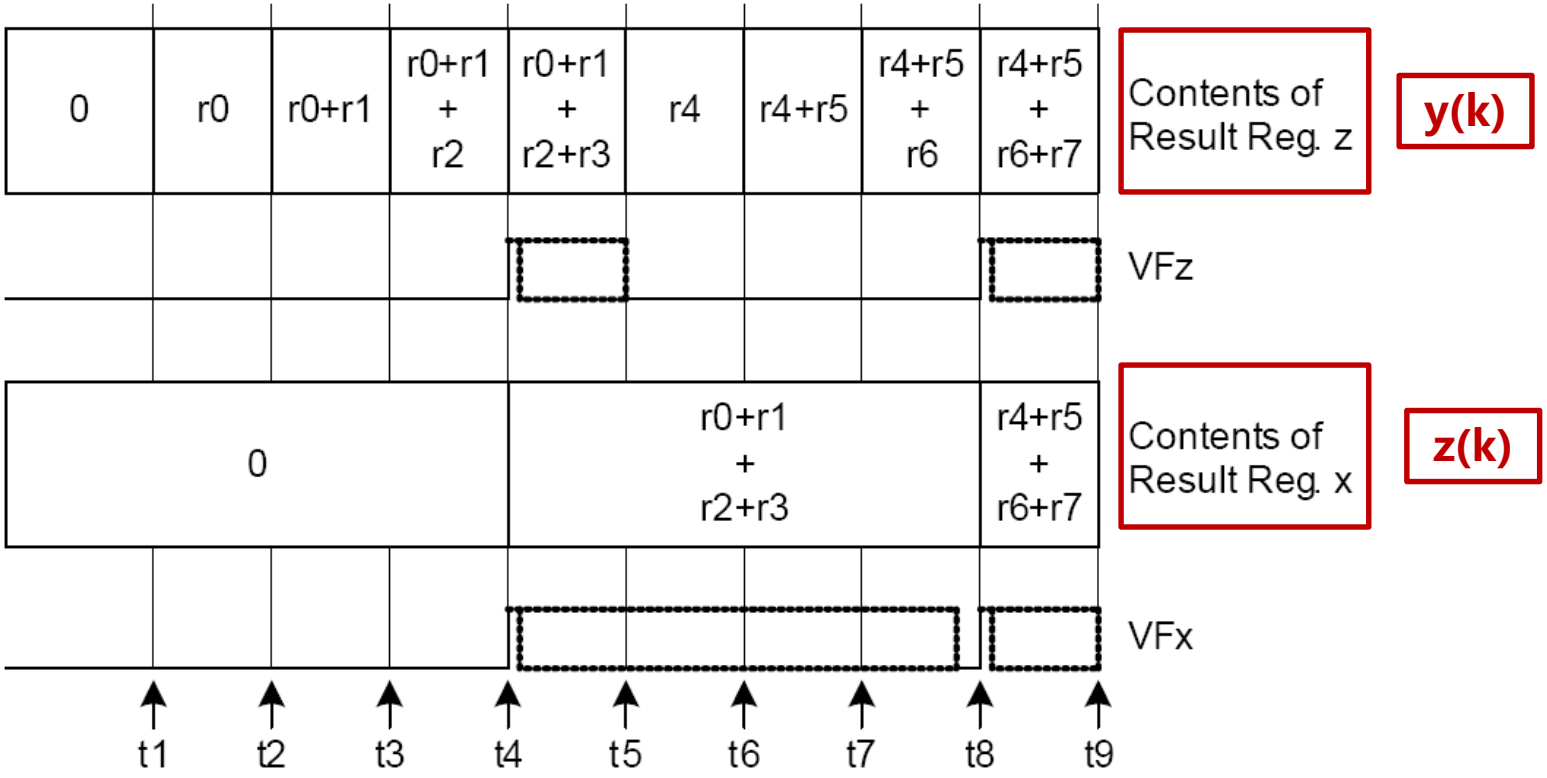
- **Data reduction mode** can be used as digital filter for anti-aliasing or decimation purposes. **DRC-Data Reduction Counter**



$$y(k)=x(k-1)+x(k-2)+x(k-3)+x(k-4)$$

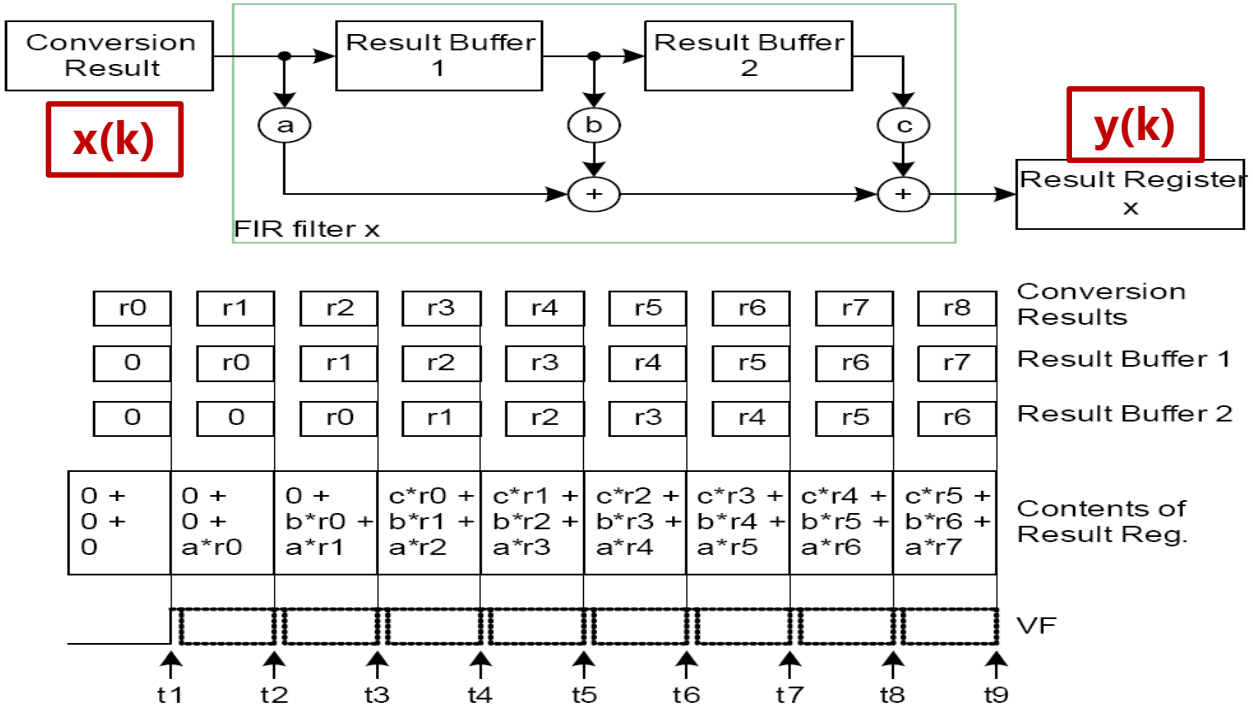
4.1.7 Conversion Result Handling
Data Modification : Standard Data Reduction Mode - FIFO

- Response time to read the final data reduction results can be increased by associating the adjacent result register to build a result FIFO.



4.1.7 Conversion Result Handling
Data Modification : Finite Impulse Response Filter Mode (FIR)

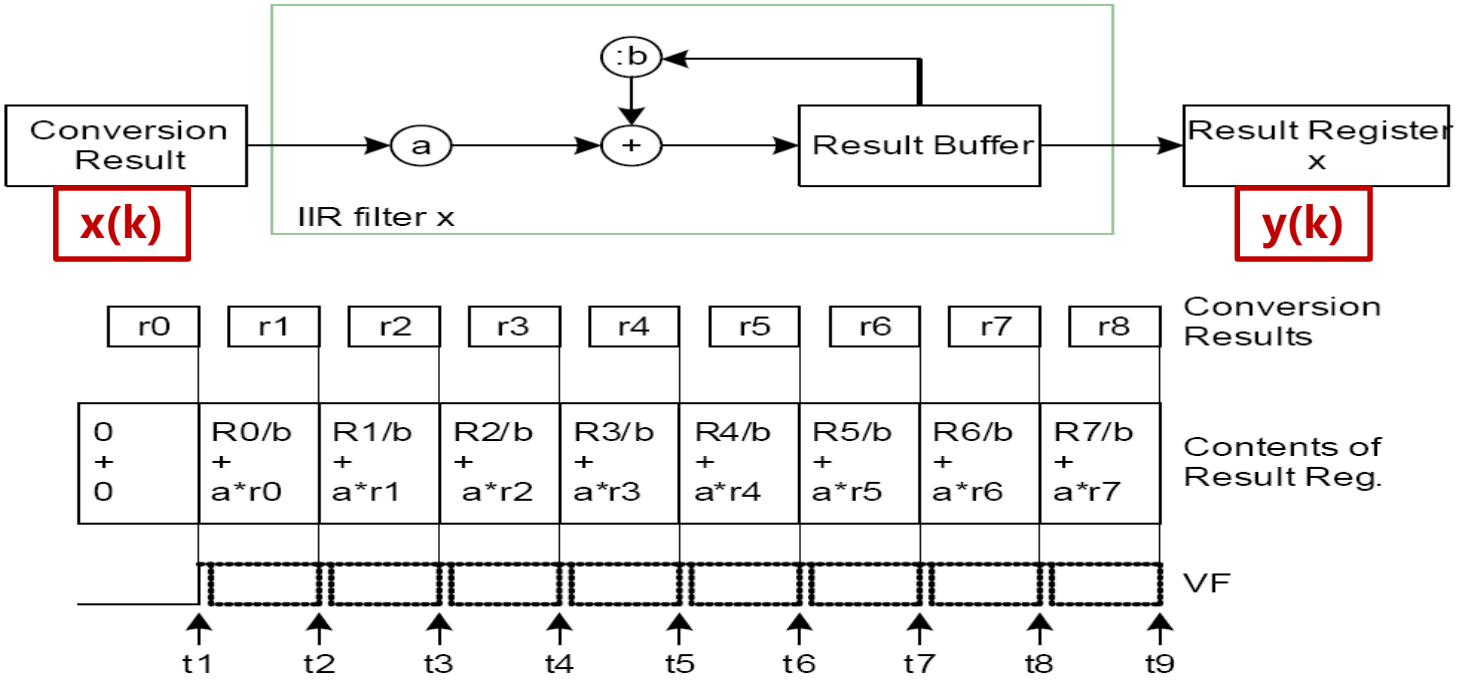
- The FIR filter uses 2 intermediate buffers (RB1, RB2) and 3 configurable coefficients (a, b, c).



$$y(k)=x(k-1)*a+x(k-2)*b+x(k-3)*c$$

4.1.7 Conversion Result Handling
Data Modification : Infinite Impulse Response Filter Mode (IIR)

- The IIR filter uses an intermediate buffer (RB) and 2 configurable coefficients (a, b),it represents **a first order low-pass filter**



$$y(k)=x(k-1)*a+y(k-1)/b$$

4.1.7 Conversion Result Handling
Data Modification：Difference mode

- Principle：**Subtracting** the contents of result **register 0** from the actual result, puts the results to the respective channel in relation to another signal

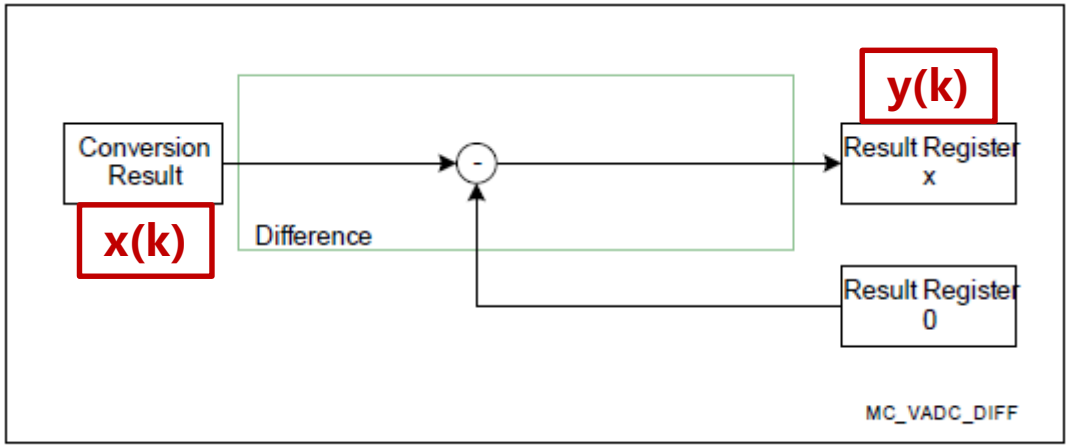


Figure 30-26 Result Difference

y(k)=x(k)-RESRegister0

4.1.8 Synchronization of Conversions
Overview

- Schedule the conversions of an ADC kernel：
  - **Synchronized Conversions for Parallel Sampling (并行采样同步转换)**
  - **Equidistant Sampling (等间距采样)**
  - **Synchronous Sampling (同步采样)**

Example

- Synchronized conversions for channel CH3 of ADC0 and ADC1
- ADC0 is the **master**, ADC1 is the **slave**

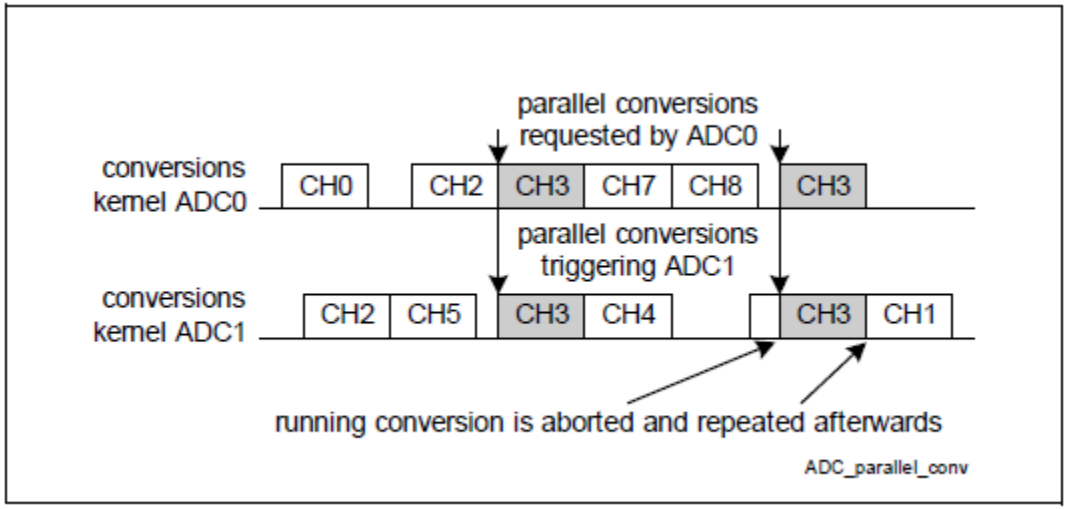


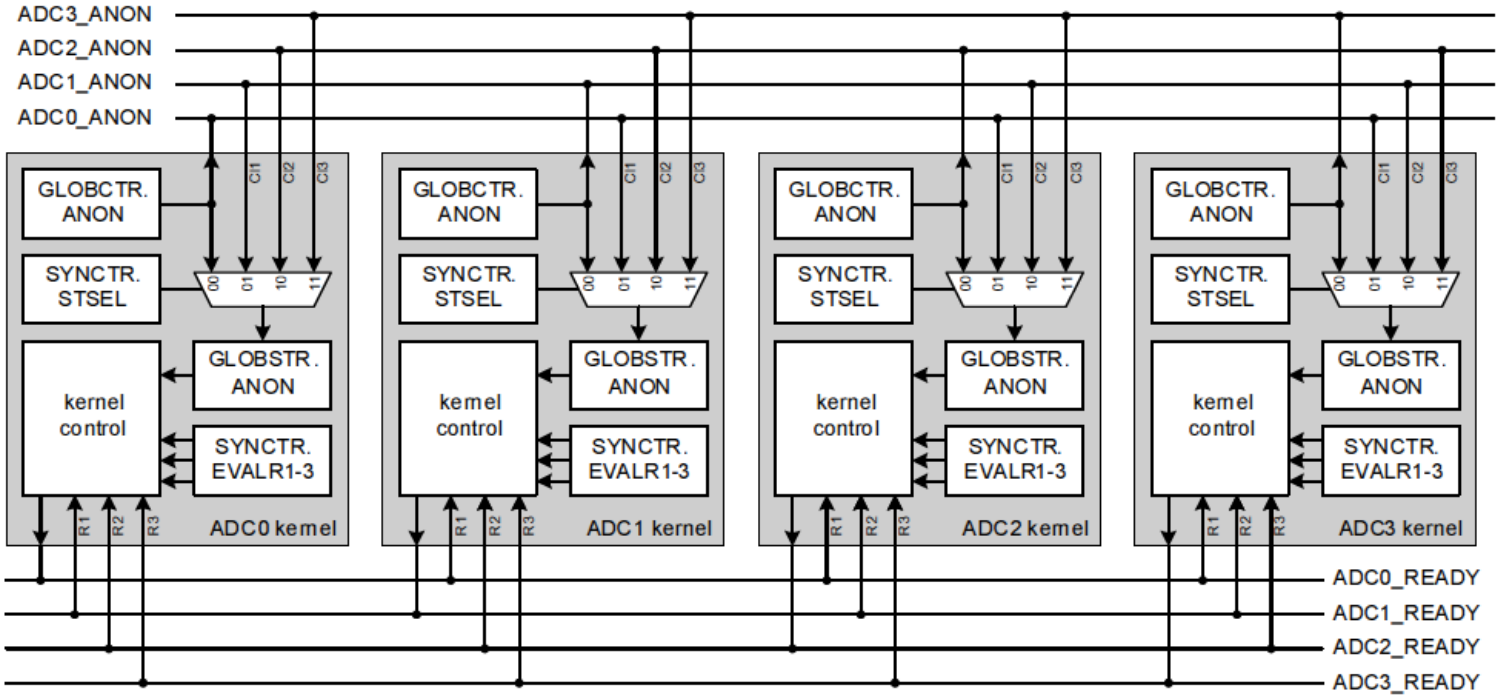
Figure 30-27 Parallel Conversions

Note : Synchronized conversions request the same channel number, defined by the master

4.1.8 Synchronization of Conversions
Synchronized Conversions for Parallel Sampling

Master and slave kernels form a "conversion group" to control parallel sampling:

- Master controls the slave by providing the control information **GxARBCFG.ANONS** and the requested **channel number**
  - Bitfields **GxSYNCTR.STSEL** select the **source of the ANON** information
- **READY** signals indicate when a kernel is ready to start the sample phase of a parallel conversion



4.1.8 Synchronization of Conversions

- The following synchronization groups are foreseen for the high-end device TC27x within the AURIX Family

Synchronization Groups in the TC27x

ADC Kernel	Synchr. Group	Master selected by control input CIx[1]			
		CI0[2]	CI1	CI2	CI3
ADC00	A	ADC00	ADC01	ADC02	ADC03
ADC01	A	ADC01	ADC00	ADC02	ADC03
ADC02	A	ADC02	ADC00	ADC01	ADC03
ADC03	A	ADC03	ADC00	ADC01	ADC02
ADC04	B	ADC04	ADC05	ADC06	ADC07
ADC05	B	ADC05	ADC04	ADC06	ADC07
ADC06	B	ADC06	ADC04	ADC05	ADC07
ADC07	B	ADC07	ADC04	ADC05	ADC06

1) The control input is selected by bitfield STSEL in register **GxSYNCTR (x = 0 - 7)**. Select the corresponding ready inputs accordingly by bits EVALRx.

2) Control input CI0 always selects the own control signals of the corresponding ADC kernel. This selection is meant for the synchronization master or for stand-alone operation.

4.1.8 Synchronization of Conversions

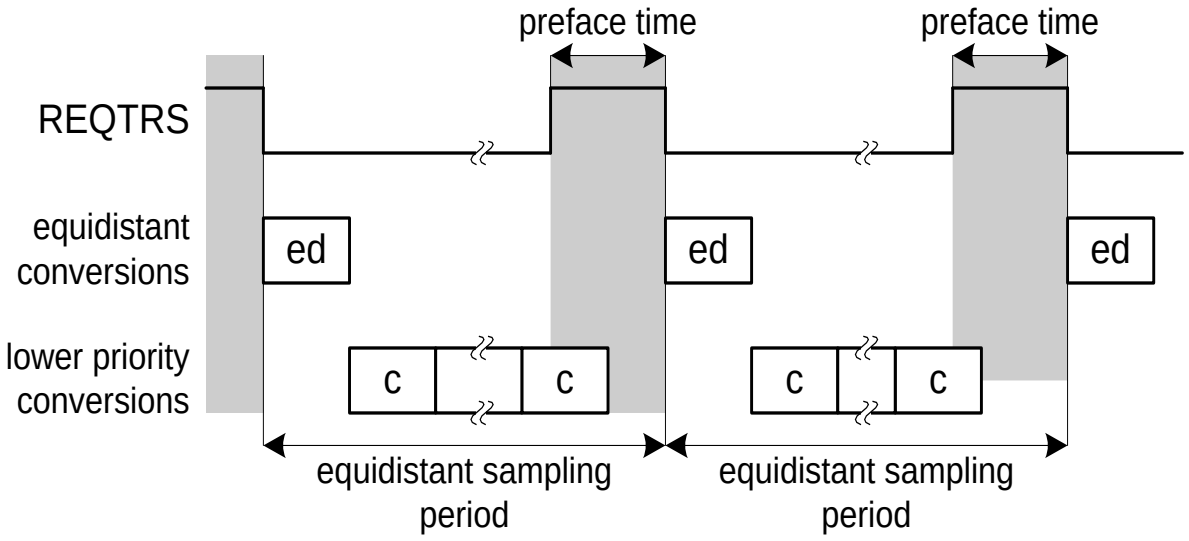
- The following converters will be implemented in the high-end device TC27x within the AURIX Family

Table 32-8 General Converter Configuration in the TC27x

Converter Group	Input Channels	Channels with Alternate Reference	12-bit Performance	Common Service Request Group
G0	0 … 7	4	Calibrated	C0
G1	0 … 7	8	Calibrated	C1
G2	0 … 7	8	Calibrated	C0
G3	0 … 7	8	Calibrated	C1
G4	0 … 7	7	Uncalibrated	C0
G5	0 … 7	8	Uncalibrated	C1
G6	0 … 7	8	Uncalibrated	C0
G7	0 … 7	8	Uncalibrated	C1

4.1.8 Synchronization of Conversions
Equidistant Sampling

- Be executed in a fixed timing raster.
- Be triggered by an external signal (e.g. a timer).
- REQTRx
  - Decide the sampling rate
  - High time means defines the preface time interval (request pending).
  - long enough for a currently active conversion to finish



4.1.8 Synchronization of Conversions
Synchronous Sampling

- Configure : **SYNSH01** and **SYNSH23** in register SHSCFG

SYNSH23, SYNSH01	[6:5], [8:7]	rw	**Synchronization of Sample&Hold Units xy**[1)] 00_B Individual sampling controlled by associated group 01_B S&H units x and y are controlled by the sample control signal of S&H unit 0 of SHS0 10_B S&H units x and y are controlled by the sample control signal of S&H unit 0 of SHS1 11_B Reserved

- Prerequisites:
  - Generate conversion requests on all involved converter groups
  - Enable synchronous sampling for the involved S&H units
  - Equal clock dividers and sample times
  - Highest priority

4.1.9 Safety Features
Overview

Several safety features :

- **Broken Wire Detection (BWD)**
  - validates the connection from the sensor to the input pin
- **Pull Down Diagnostics (PDD)**
  - check the proper connection of a signal source (sensor) to the multiplexer
- **Multiplexer Diagnostics (MD)**
  - validates the operation of the internal analog input multiplexer
- **Converter Diagnostics (CD)**
  - validates the operation of the Analog/Digital converter itself

4.1.9 Safety Features
Broken Wire Detection

- Principle : The converter's capacitor will **be pre-charged** to a **selectable** value (***V*AGND** and/or ***V*AREF**) before the regular sample phase.
  - *Note: The duration of the complete conversion is increased by the preparation phase (same as the sample phase)*

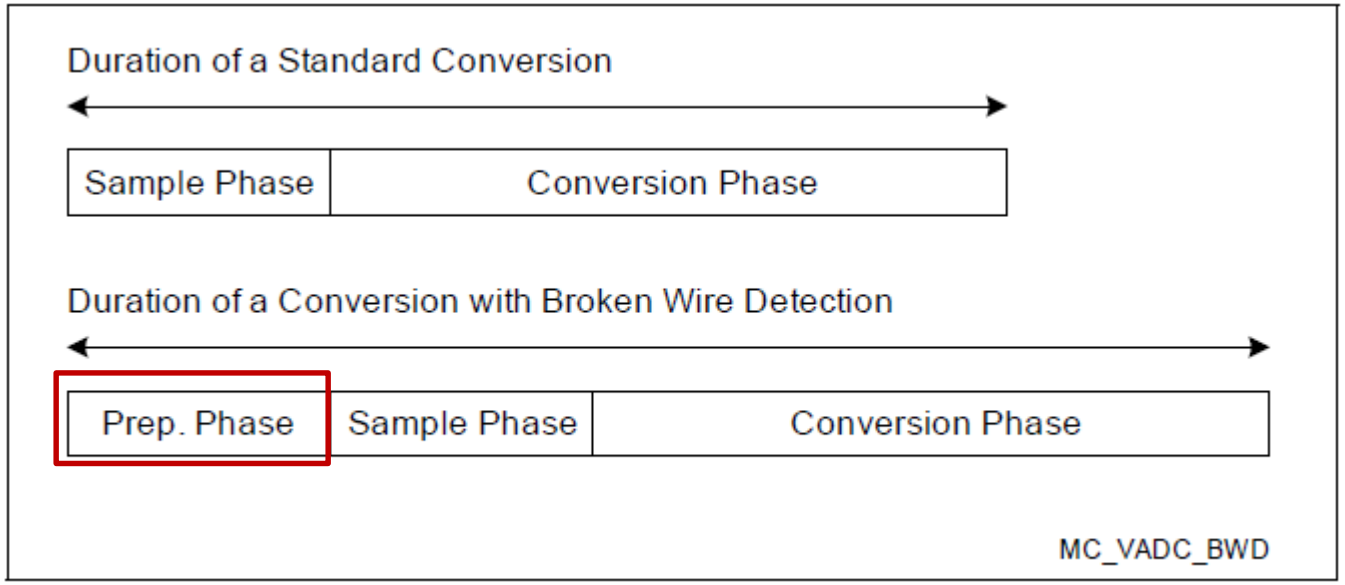


Figure 30-30 Broken Wire Detection

4.1.9 Safety Features
Signal Path Test Modes

- Additional test structures for Signal Path Test Modes.
  - **Pull Down Diagnostics (PDD)**
  - **Multiplexer Diagnostics (MD)**
  - **Converter Diagnostics (CD)**

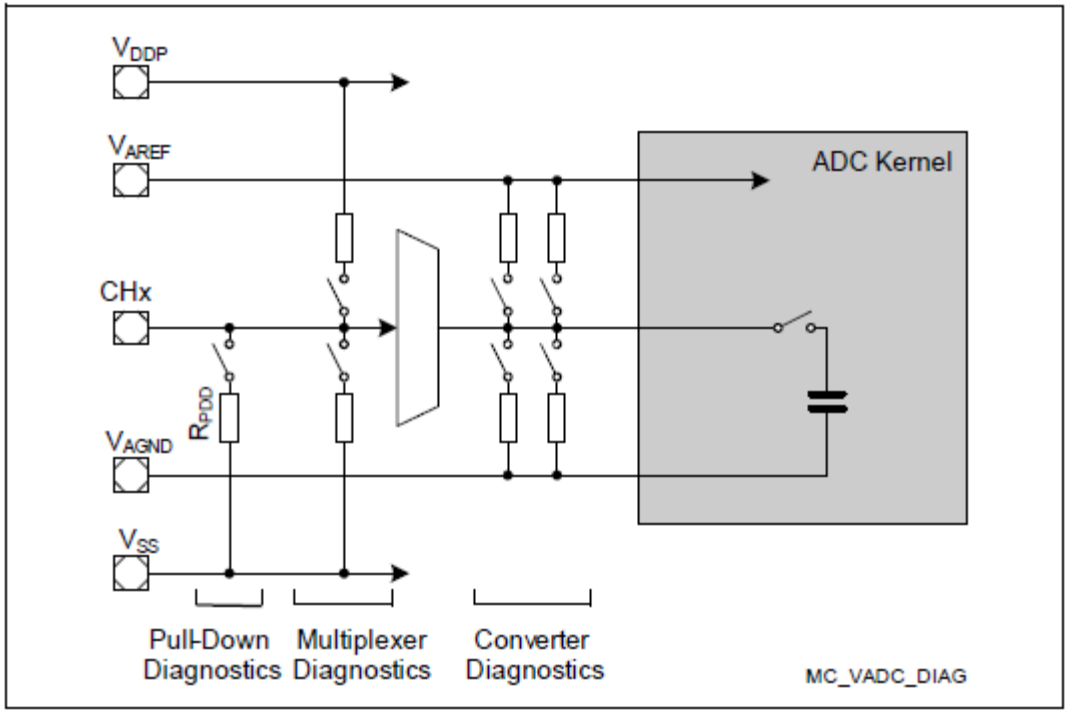


Figure 30-31 Signal Path Test

4.1.10 External Multiplexer Control
Overview

- The number of analog input channels can be increased by connecting external analog multiplexers to an input channel.
  - local filters may be used at the additional inputs

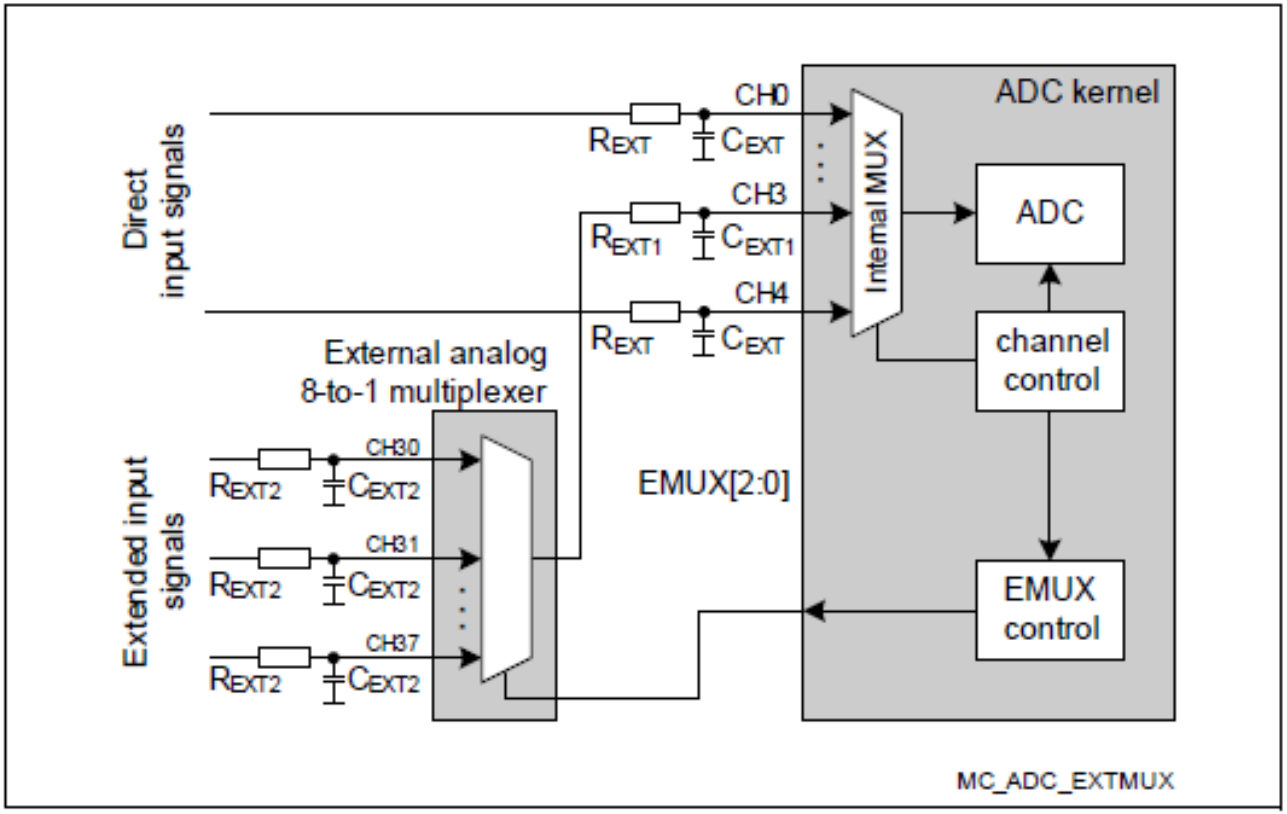


Figure 30-32 External Analog Multiplexer Example

4.1.10 External Multiplexer Control
Multiplexer Operation mode

- SAR can be configured to control these external 1-out-of-8 multiplexers automatically, arbitrary channels can be selected for this operating mode
  - **Sequence mode**
    - Conversion sequence in example: --4-**32-31-30**-2-1-0--4-**32-31-30**-2-1-0--
  - **Single-step mode**
    - Conversion sequence in example: --4-**32**-2-1-0--4-**31**-2-1-0--4-**30**-2-1-0--
  - **Steady mode**
    - Conversion sequence in example: --4-**32**-2-1-0--4-**32**-2-1-0--4-**32**-2-1-0-

4.1.10 External Multiplexer Control
Multiplexer control

- VADC use standard binary format or Gray-coded to do multiplexer control

Table 32-4 EMUX Control Signal Coding

Channel	0	1	2	3	4	5	6	7
Binary	000_B	001_B	010_B	011_B	100_B	101_B	110_B	111_B
Gray	000	001	011	010	110	111	101	100

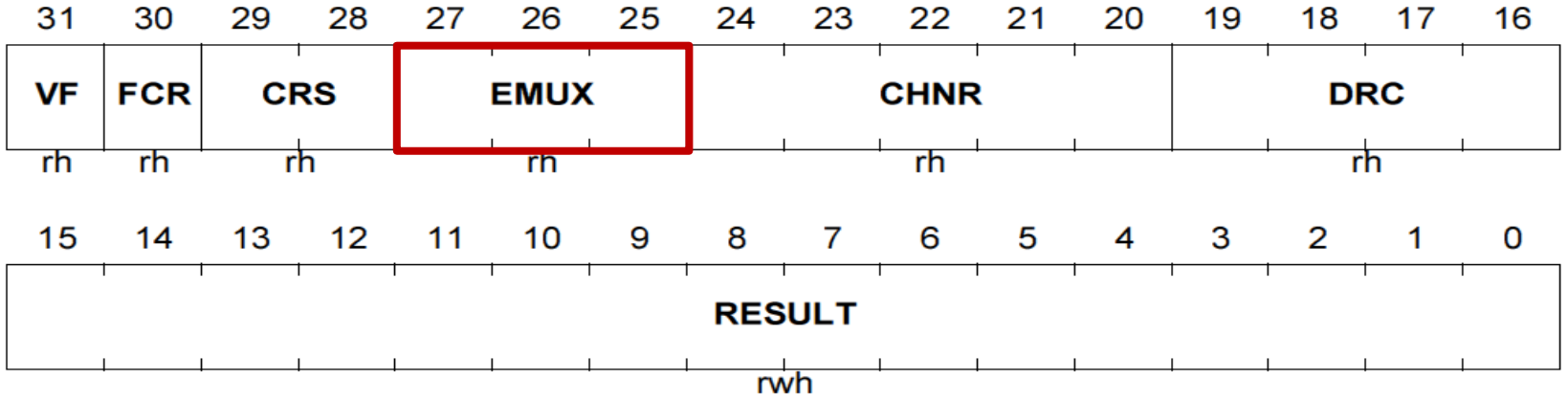
Control Signals

The external channel number that controls the external multiplexer can be output in standard binary format or Gray-coded. Gray code avoids intermediate multiplexer switching when selecting a sequence of channels, because only one bit changes at a time. **Table 27-7** indicates the resulting codes.

4.1.10 External Multiplexer Control
MUX Channel Result

- SFR GxRESy (x = 0-7), (y= 0-15)

G0RESy (y = 0 - 15)		
Group 0 Result Register y	$(0700_H + y * 0004_H)$	Reset Value: $0000\ 0000_H$
G1RESy (y = 0 - 15)		
Group 1 Result Register y	$(0B00_H + y * 0004_H)$	Reset Value: $0000\ 0000_H$
G2RESy (y = 0 - 15)		
Group 2 Result Register y	$(0F00_H + y * 0004_H)$	Reset Value: $0000\ 0000_H$
G3RESy (y = 0 - 15)		
Group 3 Result Register y	$(1300_H + y * 0004_H)$	Reset Value: $0000\ 0000_H$



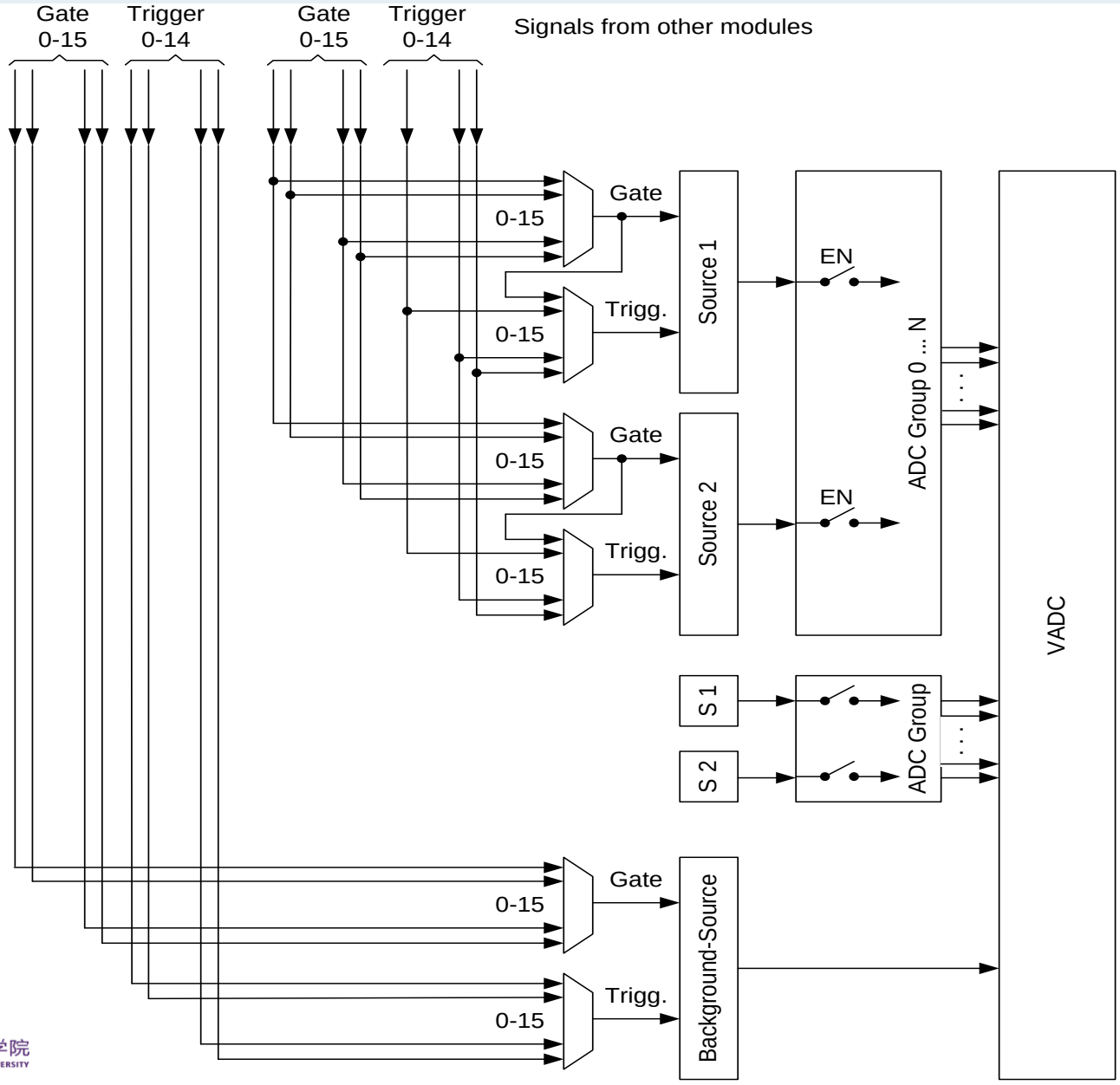
Field	Bits	Type	Description
EMUX	[27:25]	rh	**External Multiplexer Setting** Indicates the setting of the external multiplexer, corresponding to the value in bitfield RESULT. *Note: Available in GxRES0 only.* *Use GxRES0 if EMUX information is required.*

4.1.11 Service Request Generation
Overview

Each A/D Converter can activate up to **8 service request** output signals to issue a service request to CPU or DMA

- **Request source events**
- **Channel events**
- **Result events**

4.1.11 Trigger/Gate Inputs Connections
Overview



ENERGY EFFICIENCY
MOBILITY
SECURITY

Innovative semiconductor solutions for energy efficiency, mobility and security.







清华大学车辆与运载学院
SCHOOL OF VEHICLE AND MOBILITY TSINGHUA UNIVERSITY

Question

- In a arbitration round, how many arbitration slot? The second slot is for what?
- Safety feature in SADC
- Operating modes for the converter?
- Please explain the Fast Compare Mode.
- How to calculate the conversion time for the compatible time mode?
- Please explain the properties of 10-Bit Right-Aligned method for both standard and accumulated conversions.
- Understand the four Data Reduction Modes.
- Please explain the three multiplexer operation modes for external multiplexer control using examples?

本节结束，谢谢！

Thank you！